MTH2021 Mathematical Methods 2
Part 2: Partial Differential Equations

Lecture Notes 2023/2024

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Contents

Preface

> If you assume continuity, you can open the well-stocked mathematical toolkit of continuous functions and differential equations, the saws and hammers of engineering and physics for the past two centuries (and the foreseeable future).
>
> *The (Mis)Behaviour of Markets: A Fractal View of Risk, Ruin and Reward*
> Benoit Mandelbrot and Richard Hudson

The second half of Mathematical Methods 2 provides an introduction to Partial Differential Equations (PDEs). It explains what PDEs are, shows how the most common PDEs are derived, and presents a few methods with which they can be solved, including Fourier series and Fourier integrals. It also shows that solving PDEs may require us to introduce new functions, such as the Bessel functions. These and other such functions are known as the *special functions of mathematical physics*, because the equations that they solve emerged in physics in an effort to describe and understand the world around us. The main equations of Maxwell's Electromagnetic Theory (1865), Einstein's General Relativity (1915) and Quantum Theory (Heisenberg and Schrödinger, 1926), are all partial differential equations. They also describe the transfer of heat and the flow of liquids and gases (Fluid Dynamics).

However, the applicability and importance of partial differential equations is by no means confined to physics. They are also used widely in biology, weather modelling, economics, engineering, etc.

These notes are self-contained and do not follow any particular book. However, students should be familiar with elementary calculus (derivatives, integrals, simple ordinary differential equations) and calculus of functions of several variables ("Calculus in 3D"), including partial derivatives and the main ideas of the vector field theory, as presented in MTH1021 Mathematical Methods 1.

Numerous texts on PDEs are available, and only a small subset of these is listed below, with their QUB Library shelfmarks. These books contain much more than what we are going to cover in MTH2021, but can be used for more examples, alternative explanations and further directions of study.

1. N. H. Asmar, Partial Differential Equations with Fourier Series and Boundary Value Problems (Pearson Prentice Hall, Upper Saddle River, NJ, 2005) QA577/ASMA.

2. P. W. Berg and J. L. McGregor, Elementary Partial Differential Equations (Holden-Day, San Francisco, 1966) QA374/BERG.

3. A. V. Bitsadze and D. F. Kalinichenko, A Collection of Problems on the Equations of Mathematical Physics (Mir, Moscow, 1980) QC20/BITS.

4. C. Constanda, Solution techniques for elementary partial differential equations (CRC Press, Boca Raton, 2010) QA377/CONS.

5. R. Courant and D. Hilbert, Methods of Mathematical Physics, Vols. 1 and 2 (Wiley, New York, 1989) QA401/COUR.

6. R. Haberman, Applied Partial Differential Equations: with Fourier Series and Boundary Value Problems (Prentice Hall, Harlow, 2004) QA377/HABE.

7. Y. Pinchover and J. Rubinstein, An Introduction to Partial Differential Equations (Cambridge University Press, Cambridge, 2005) QA374/PINC.

8. G. Stephenson, Partial Differential Equations for Scientists and Engineers (Longman, London, 1985) QA374/STEP.

9. W. A. Strauss, Partial Differential Equations: an Introduction (Wiley, New York, 1992) QA374/STRA.

10. H. F. Weinberger, A First Course in Partial Differential Equations, with Complex Variables and Transform Methods (John Wiley, London, 1965) QA374/WEIN.

1 Introduction. Examples of partial differential equations

1.1 Terminology

An equation that links a function and its derivatives is a *differential equation*.

If the function depends on only one independent variable, this is an *ordinary* differential equation (ODE).

If the function depends on several (two or more) independent variables, this is a *partial* differential equation (PDE).

A general PDE is a relation of the form

$$F(x, y, \dots, u, u_x, u_y, \dots, u_{xx}, u_{xy}, \dots) = 0, \tag{1.1}$$

where F is a function of variables x, y, $\dots$, u, u_x, u_y, $\dots$, u_{xx}, u_{xy} $\dots$. A function $u(x, y, \dots)$ of the *independent variables* x, y, $\dots$, which identically satisfies Eq. (1.1), is a *solution* of the PDE.

Here we use the following compact notation for the partial derivatives:

$$u_x = \frac{\partial u}{\partial x}, \qquad u_y = \frac{\partial u}{\partial y}, \qquad \dots \tag{1.2}$$

$$u_{xx} = \frac{\partial^2 u}{\partial x^2}, \quad u_{xy} = \frac{\partial^2 u}{\partial x \partial y}, \quad \dots. \tag{1.3}$$

The *order* of a PDE is the order of the highest derivative it contains.

A PDE is *linear* if F is linear in u, u_x, u_y, $\dots$, u_{xx}, u_{xy}, $\dots$, with coefficients depending only on the independent variables x, y, $\dots$. Otherwise the equation is called *nonlinear*.

A linear PDE is *homogeneous* if

$$F(x, y, \dots, Cu, Cu_x, \dots, Cu_{xx}, \dots) = CF(x, y, \dots, u, u_x, \dots, u_{xx}, \dots).$$

Otherwise, it is *inhomogeneous*.

Examples.

1. $\dfrac{du}{dx} + p(x)u = q(x)$ for $u(x)$ is a 1st-order linear inhomogeneous ordinary differential equation (ODE); for $q = 0$ it is homogeneous.[1]
2. $au'' + bu' + cu = 0$ for $u(x)$ is a 2nd-order linear homogeneous ODE with constant coefficients; $au'' + bu' + cu = f(x)$ is inhomogeneous.[1]
3. $xy\dfrac{\partial u}{\partial x} + u\dfrac{\partial u}{\partial y} = x^2$ for $u(x, y)$ is a 1st-order, nonlinear PDE.

[1] See MTH1021 Mathematical Methods 1 for methods of solving these equations.

4. $\dfrac{\partial^2 u}{\partial x^2} - xy\dfrac{\partial u}{\partial y} = 0$ for $u(x,y)$ is 2nd-order linear homogeneous.

5. $\dfrac{\partial^4 u}{\partial x^4} + \dfrac{1}{a^4}\dfrac{\partial^2 u}{\partial t^2} = q(x,t)$ for $u(x,t)$ is 4th-order linear inhomogeneous.

   [It describes the shape of a beam under normal stress $q(x,t)$.]

6. $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} + \dfrac{\partial^2 u}{\partial z^2} = 0$ for $u(x,y,z)$ is 2nd-order linear homogeneous.

   This is Laplace's equation. It is important in electrostatics, fluid dynamics, etc. It can be written in compact form as $\nabla^2 u = 0$, where

   $$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}, \tag{1.4}$$

   is the Laplacian.[2] The inhomogeneous version of Laplace's equation,

   $$\nabla^2 u = f(x,y,z)$$

   is known as Poisson's equation.

1.2 Variety of solutions

"General solutions" contain arbitrary functions with one independent variable less than u.

Examples of equations and solutions.

1. Equation: $u_y = 0$ for $u(x,y)$.

   Solution: $u = w(x)$, where w is an arbitrary function.

2. Equation: $u_{xy} = 0$ for $u(x,y)$.

   Solution: $u = v(x) + w(y)$, where v and w are arbitrary functions.

3. Equation: $u_{xy} = f(x,y)$ for $u(x,y)$.

   Solution: $u = \int_{x_0}^{x}\int_{y_0}^{y} f(x,y)dydx + v(x) + w(y)$.

4. Equation: $u_x = u_y$ for $u(x,y)$.

   Using new variables, $\xi = x+y$, $\eta = x-y$,

   $$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi}\frac{\partial \xi}{\partial x} + \frac{\partial u}{\partial \eta}\frac{\partial \eta}{\partial x} = \frac{\partial u}{\partial \xi} + \frac{\partial u}{\partial \eta},$$

   $$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial \xi}\frac{\partial \xi}{\partial y} + \frac{\partial u}{\partial \eta}\frac{\partial \eta}{\partial y} = \frac{\partial u}{\partial \xi} - \frac{\partial u}{\partial \eta},$$

   and the equation is reduced to $u_\eta = 0$.

   Solution: $u = w(\xi)$, or $u = w(x+y)$, where w is an arbitrary function.

[2] Recall the basics of the vector field theory: the nabla operator, $\nabla = \mathbf{i}\frac{\partial}{\partial x} + \mathbf{j}\frac{\partial}{\partial y} + \mathbf{k}\frac{\partial}{\partial z}$, where $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are unit vectors along the x, y and z axes; $\nabla\cdot\nabla = \nabla^2$ is the Laplacian; $\nabla u = \operatorname{grad} u$, $\nabla\cdot\mathbf{A} = \operatorname{div}\mathbf{A}$, $\nabla\times\mathbf{A} = \operatorname{curl}\mathbf{A}$ (introduced in MTH1021).

5. Equation: $u_{xx} - u_{yy} = 0$ for $u(x, y)$.

   The equation can be written as

$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y}\right)\left(\frac{\partial}{\partial x} - \frac{\partial}{\partial y}\right) u = 0.$$

   From example 4 above, using $\xi = x + y$ and $\eta = x - y$,

$$\frac{\partial}{\partial x} + \frac{\partial}{\partial y} = 2\frac{\partial}{\partial \xi}, \quad \frac{\partial}{\partial x} - \frac{\partial}{\partial y} = 2\frac{\partial}{\partial \eta},$$

   and the differential equation becomes

$$4\frac{\partial}{\partial \xi}\frac{\partial}{\partial \eta} u = 0.$$

   From example 2, we have $u = w(\xi) + v(\eta)$.

   Solution: $u = w(x+y) + v(x-y)$, where v and w are arbitrary functions.

Exercise. Show that the solution of $u_{tt} - c^2 u_{xx} = 0$ for $u(x, t)$ is

$$u = w(x + ct) + v(x - ct),$$

where v and w are arbitrary functions. This is known as *d'Alembert's solution* of the wave equation (see Sec. 1.3).

[Hint: use new variables $\xi = x + ct$ and $\eta = x - ct$.]

1.3 Wave equation

Let us derive the equation of motion for a string with linear mass density (i.e., mass per unit length) ρ under tension T. In equilibrium the string is along the x axis, Fig. 1(a). Let us consider the motion of a small segment dx of the string with mass $dm = \rho dx$. Let $u(x, t)$ be the displacement of the string at point x in the direction perpendicular to the x axis. Away from equilibrium, this segment is acted upon by two forces, $T(x)$ and $T(x + dx)$, which make angles $\theta(x)$ and $\theta(x + dx)$ with the x axis, Fig. 1(b).

Newton's second law ($ma = F$) for the motion of the segment dx in the direction perpendicular to the x axis reads

$$dm \frac{\partial^2 u}{\partial t^2} = T(x + dx)\sin\theta(x + dx) - T(x)\sin\theta(x), \tag{1.5}$$

where we have taken the perpendicular components of the tension forces. The right-hand-side is the change in $T(x)\sin\theta(x)$ over dx, hence[3]

$$\rho dx \frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial x}\left[T(x)\sin\theta(x)\right] dx. \tag{1.6}$$

The right-hand-side can be simplified if we consider *small* vibrations of the string. In this case the length of the strength remains practically unchanged, so that

$$T(x) \approx \text{const} \equiv T, \tag{1.7}$$

[3] Recall the basic relations $df = f'(x)dx$ and $df \simeq f(x + dx) - f(x)$.

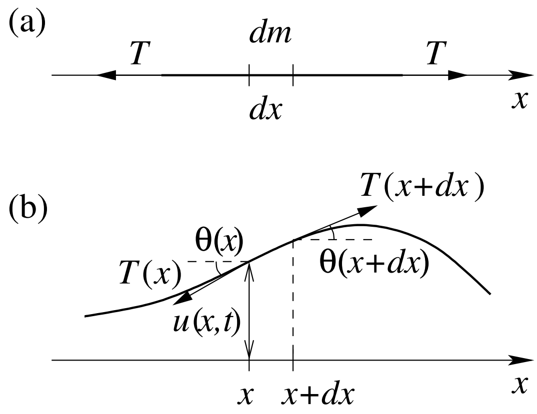


Figure 1: (a) shows the string in equilibrium; (b) shows the forces acting on the string segment between points x and $x + dx$, away from equilibrium.

and the angle the string makes with the x axis is small, $\theta \ll 1$, so that[4]

$$\sin\theta \approx \tan\theta = \frac{\partial u}{\partial x}. \tag{1.8}$$

Using (1.7) and (1.8) in Eq. (1.6), and cancelling dx, we obtain

$$\rho\,\frac{\partial^2 u}{\partial t^2} = T\frac{\partial^2 u}{\partial x^2}. \tag{1.9}$$

Dividing by ρ, regrouping and introducing

$$c = \sqrt{\frac{T}{\rho}}, \tag{1.10}$$

we obtain

$$\frac{\partial^2 u}{\partial t^2} - c^2\frac{\partial^2 u}{\partial x^2} = 0, \tag{1.11}$$

which is known as the *wave equation in one dimension*. This 2nd-order linear homogeneous PDE can also be written in compact form, as

$$u_{tt} - c^2 u_{xx} = 0. \tag{1.12}$$

The meaning of the constant c becomes clearer if we analyse its dimension. Using the dimensions of the force and linear mass density, $[T] = \mathrm{kg\,m/s^2}$ and $[\rho] = \mathrm{kg/m}$ (in SI units), we see that the dimension of c,

$$[c] = \sqrt{\frac{\mathrm{kg\,m^2}}{\mathrm{s^2\,kg}}} = \frac{\mathrm{m}}{\mathrm{s}}, \tag{1.13}$$

is that of *speed*. As we will see later, solutions of Eq. (1.11) describe waves propagating along the string with speed c.

[4] Recall the Maclaurin expansions, $\sin\theta = \theta - \frac{1}{6}\theta^3 + \ldots$, $\cos\theta = 1 - \frac{1}{2}\theta^2 + \ldots$, which for $\theta \ll 1$ give $\sin\theta \simeq \theta$ and $\cos\theta \simeq 1$, so that $\tan\theta = \sin\theta/\cos\theta \simeq \theta$.

If an external force $f(x,t)$ (per unit length) acts perpendicular to the string,[5] then $f(x,t)dx$ must be added to the right-hand-side of Newton's second law (1.5). Then instead of Eq. (1.11) we obtain the following *inhomogeneous* equation:

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = \frac{f(x,t)}{\rho}. \tag{1.14}$$

In particular, by finding the time-independent solution of this equation (i.e., such that $\partial u/\partial t = 0$), one can determine the equilibrium shape of the string under gravity.

The wave equation in two dimensions (membrane)

$$\frac{\partial^2 u}{\partial t^2} - c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) = 0, \tag{1.15}$$

and in three dimensions (sound in air, electromagnetic waves),

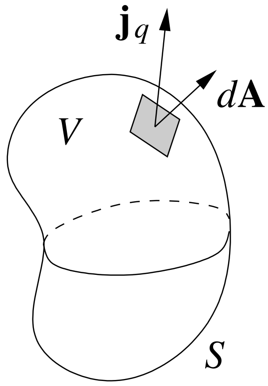
$$\frac{\partial^2 u}{\partial t^2} - c^2 \nabla^2 u = 0. \tag{1.16}$$

1.4 Heat equation

Fourier's law of heat conduction states:

$$\mathbf{j}_q = -\kappa \nabla T, \tag{1.17}$$

where $\mathbf{j_q}$ is the heat flux density (i.e., the amount of heat energy which flows through a unit area perpendicular to $\mathbf{j}_q$ in unit time), T is the temperature, and κ is the thermal conductivity of the medium. The gradient vector ∇T is in the direction of the fastest growth in T. The minus sign in Eq. (1.17) indicates that heat flows in the opposite direction, i.e., towards lower temperatures.



Let us first find the mathematical form of the heat energy conservation. Consider an arbitrary volume V bounded by surface S. Let $q(\mathbf{r},t)$ be the heat energy density (i.e., the heat energy per unit volume at point $\mathbf{r}$ and time t). The total heat energy in V is given by the integral is over the volume V:

$$\int_V q dV. \tag{1.18}$$

The energy inside V can change only due to the heat flowing through S. Consider a small surface element $d\mathbf{A}$, where $|d\mathbf{A}|$ is its area, and $d\mathbf{A}$ is in the direction of the outward normal to the surface. The amount of heat energy which flows out of V through this element in time dt is $\mathbf{j}_q \cdot d\mathbf{A}dt$, where $\mathbf{j}_q$ is the heat flux density at this point on the surface.[6] Hence, we can relate the *change* in the heat energy in V to the total energy flux through the surface,

$$d \int_V q dV = - \oint_S \mathbf{j}_q \cdot d\mathbf{A}\, dt. \tag{1.19}$$

[5] For example, if the string is horizontal and the external force is gravity, then $f(x,t) = \rho g$, where g is the acceleration due to gravity, and the positive u direction is downwards.

[6] The scalar product here ensures that only the component of $\mathbf{j}_q$ parallel to $d\mathbf{A}$ contributes (e.g., if $\mathbf{j}_q$ is perpendicular to $d\mathbf{A}$ then $\mathbf{j}_q \cdot d\mathbf{A} = 0$, the heat flowing neither in nor out, but parallel to the surface.

The minus sign here relates to the choice of direction of $d\mathbf{A}$; if the heat flows out of V, the energy inside v is *reduced*.

Dividing Eq. (1.19) by dt and using Gauss's theorem for the integral on the right-hand-side,

$$\oint_S \mathbf{j}_q \cdot d\mathbf{A} = \int_V \nabla \cdot \mathbf{j}_q \, dV, \tag{1.20}$$

we have

$$\frac{d}{dt}\int_V q dV = -\int_V \nabla \cdot \mathbf{j}_q \, dV. \tag{1.21}$$

Swapping differentiation with respect to time and integration over the spatial coordinates, we obtain

$$\int_V \left(\frac{\partial q}{\partial t} + \nabla \cdot \mathbf{j}_q\right) dV = 0. \tag{1.22}$$

Since this relations hold for *any* volume V, we must have

$$\frac{\partial q}{\partial t} + \nabla \cdot \mathbf{j}_q = 0. \tag{1.23}$$

This equation is the differential form of the heat energy conservation.

The heat energy density is proportional to the temperature, $q = c\rho T$, where c is the specific heat and ρ is the density of the medium.

Using Fourier's law (1.17) in Eq. (1.23) gives

$$\frac{\partial q}{\partial t} - \kappa \nabla \cdot \nabla T = 0. \tag{1.24}$$

The heat energy density is proportional to the temperature, $q = c\rho T$, where c is the specific heat[7] and ρ is the density of the medium. Substituting this into (1.24) and assuming that c and ρ are constant, we obtain the *heat equation*:

$$\frac{\partial T}{\partial t} - K\nabla^2 T = 0, \tag{1.25}$$

where $K = \kappa/c\rho$ is the parameter characteristic of the medium.

If we use u instead of T for the function, the heat equation takes the form

$$u_t - K\nabla^2 u = 0. \tag{1.26}$$

For a uniform rod along the x axis, the heat equation reads

$$u_t - K u_{xx} = 0. \tag{1.27}$$

[7] Specific heat is the amount of energy required to increase the temperature of a unit mass by one degree. For example, for water, iron and gold $c = 4\,200$, 460, and 130 $\mathrm{J\,kg^{-1}K^{-1}}$, respectively. Here J is joule and K is kelvin, the unit of temperature equal to the Celsius (centigrade) degree.

For a stationary temperature distribution, $\partial u/\partial t = 0$, Eq. (1.26) becomes the Laplace equation,

$$\nabla^2 u = 0. \tag{1.28}$$

This equation is also obeyed by the electrostatic potential in the region of space with no charges.

If heat energy is produced by external sources at the rate $Q(\mathbf{r},t)$ per unit volume, an extra term, $\int_V Q dV\, dt$, must be added to the right-hand side of the energy balance equation (1.19). As a result, the energy conservation (1.23) will read,

$$\frac{\partial q}{\partial t} + \nabla \cdot \mathbf{j}_q = Q(\mathbf{r},t), \tag{1.29}$$

and the heat equation (1.26) will change to

$$u_t - K\nabla^2 u = f(\mathbf{r},t), \tag{1.30}$$

where $f(\mathbf{r},t) = Q(\mathbf{r},t)/c\rho$. This is an inhomogeneous version of the linear equation (1.26). The right-hand of (1.30) is known as the *source term*.

An equation similar to (1.26) describes *diffusion* of a substance in a medium,[8]

$$\frac{\partial u}{\partial t} - D\nabla^2 u = 0, \tag{1.31}$$

where D is the diffusion coefficient, and u is the number density (or concentration) of the substance.

Another similar equation is the Schrödinger equation for the wavefunction $\psi(x,y,z,t)$ of a particle in quantum mechanics,

$$i\hbar\frac{\partial \psi}{\partial t} + \left[\frac{\hbar^2}{2m}\nabla^2\psi - U(x,y,z)\psi\right] = 0, \tag{1.32}$$

where $\hbar$ is Planck's constant, m is the mass of the particle, and $U(x,y,z)$ is its potential energy. (1.32) looks like a diffusion equation in imaginary time.

1.5 Initial and boundary conditions

1. Wave equation for the string of length l.
   (a) Boundary conditions: $u(0,t) = 0$, $u(l,t) = 0$ (fixed ends). The motion of the string will be determined if the initial conditions are specified: $u(x,0) = f(x)$ (displacement), $u_t(x,0) = g(x)$ (velocity).
   (b) If one of the ends of the string is free (e.g., at $x = l$), then $u_x(l,t) = 0$ (the *natural* boundary condition).
   (c) Boundary conditions for a string with a fixed and a driven end: $u(0,t) = 0$, $u(l,t) = h(t)$.
2. Heat equation in one dimension (rod of length l).

[8] For example, a perfume in air, or ink in water.

(a) Initial condition: $u(x,0) = f(x)$ (temperature at $t=0$). Boundary conditions: $u(0,t) = T_1$, $u(l,t) = T_2$ (ends are kept at fixed temperature).

(b) Rod with insulated ends: $u_x(0,t) = 0$, $u_x(l,t) = 0$.

(c) Newton's law of cooling: the heat flux is proportional to difference between the temperature of the body and ambient temperature. If the temperatures near the two ends of the rod are T_1 and T_2, the boundary conditions are: $-\kappa u_x(0,t) = \alpha_1[T_1 - u(0,t)]$, $-\kappa u_x(l,t) = \alpha_2[u(l,t) - T_2]$.

1.6 Linear differential equations

A linear homogeneous differential expression for a function $u(x,y,\dots)$ is

$$L[u] = Au + Bu_x + Cu_y + \cdots + Du_{xx} + Eu_{xy} + \dots, \tag{1.33}$$

where

$$L = A + B\frac{\partial}{\partial x} + C\frac{\partial}{\partial y} + \cdots + D\frac{\partial^2}{\partial x^2} + E\frac{\partial^2}{\partial x \partial y} + \dots . \tag{1.34}$$

is a linear differential operator, for which

$$L[c_1 u_1 + c_2 u_2] = c_1 L[u_1] + c_2 L[u_2]. \tag{1.35}$$

The general linear differential equation has the form

$$L[u] = f(x,y,\dots). \tag{1.36}$$

If $f = 0$, the equation is homogeneous, $f \neq 0$ – inhomogeneous.

Superposition principle: if u_1, u_2, $\dots$ are solutions of the homogeneous equation, then $c_1 u_1 + c_2 u_2 + \dots$ is also a solution.

If u_p is a solution of (1.36) with $f \neq 0$, other solutions of the inhomogeneous equation can be constructed as $u = u_p + c_1 u_1 + c_2 u_2 + \dots$.

Boundary conditions can also be homogeneous, e.g., $u(0,t) = 0$, $u_x(0,t) = 0$, or $\alpha u(0,t) + \beta u_x(0,t) = 0$, or inhomogeneous: $u(0,t) = T$, $u(l,t) = h(t)$, etc.

2 The method of variable separation

2.1 Variable separation in partial differential equations

Particular solutions of a linear homogeneous equation,

$$L[u] = 0, \tag{2.1}$$

in n independent variables $x,\ y, \dots$, can sometimes be found in the form

$$u(x, y, \dots) = X(x)Y(y)\dots\ . \tag{2.2}$$

Variable separation involves substituting (2.2) into (2.1), introducing $n-1$ *separation constants* $k_1, k_2, \dots, k_{n-1}$, and solving n ordinary differential equations for $X(x)$, $Y(y)$, $\dots$.

These solutions can be combined using the superposition principle to form a more general family of solutions (subject to certain boundary conditions).

2.2 Variable separation for the wave equation for a string

We seek solution of the wave equation for a string of length l with fixed ends, $u(0,t) = u(l,t) = 0$, in the form

$$u(x,t) = v(x)q(t). \tag{2.3}$$

Substituting this into (1.11) yields

$$\frac{\partial^2[v(x)q(t)]}{\partial t^2} - c^2\frac{\partial^2[v(x)q(t)]}{\partial x^2} = 0,$$

or

$$v(x)\frac{d^2q(t)}{dt^2} - c^2q(t)\frac{d^2v(x)}{dx^2} = 0, \tag{2.4}$$

where we have replaced the partial derivatives by the "total" ones (using d instead of ∂), as they now act on functions of one variable (either t or x). Dividing Eq. (2.4) through by $u(x,t) = v(x)q(t)$ are rearranging, we obtain

$$\underbrace{\frac{1}{q(t)}\frac{d^2q}{dt^2}}_{\lambda} - c^2\underbrace{\frac{1}{v(x)}\frac{d^2v}{dx^2}}_{\lambda} = 0. \tag{2.5}$$

In this equation, the first term depends only on t, while the second term – only on x. For the equality to hold *for all* x and t, each term must be equal to a constant (known as the separation constant). Denoting this constant by λ and assuming that it is negative[9], $\lambda = -\omega^2$, we get from the first term:

$$\frac{d^2q}{dt^2} + \omega^2q = 0. \tag{2.6}$$

[9] A positive or zero separation constant does not allow one to satisfy the boundary conditions $u(0,t) = u(l,t) = 0$, see below.

The general general solution of this equation is

$$q(t) = C_1 \cos \omega t + C_2 \sin \omega t, \tag{2.7}$$

where C_1 and C_2 are arbitrary constants. It can also be written as

$$q(t) = C \cos(\omega t + \alpha), \tag{2.8}$$

where $C > 0$ and α is arbitrary.[10] Thus we see that $q(t)$ undergoes harmonic oscillations with the (angular) frequency ω. The period of such oscillations is $T = 2\pi/\omega$, while C in Eq. (2.8) is their *amplitude* and α is the *initial phase*.[11]

From the second term in Eq. (2.5), using $\lambda = -\omega^2$, we have

$$c^2 \frac{d^2 v}{dx^2} = -\omega^2 v(x),$$

which can be written as

$$\frac{d^2 v}{dx^2} + k^2 v = 0, \tag{2.9}$$

where $k = \omega/c > 0$. The general solution of this equation is

$$v(x) = A \cos kx + B \sin kx, \tag{2.10}$$

where A and B are arbitrary constants. This solution is periodic in x, with the period $2\pi/k$ known as the wavelength, while k is called the *wavenumber*.

For $u(x,t) = v(x)q(t)$ to satisfy the boundary conditions $u(0,t) = u(l,t) = 0$ for all t, we must have $v(0) = 0$ and $v(l) = 0$. Applying the first of these to $v(x)$ in Eq. (2.10) gives $A = 0$, so that $v(x) = B \sin kx$. The second boundary condition then gives

$$B \sin kl = 0.$$

A nonzero $v(x)$ requires $B \neq 0$. Hence, $\sin kl = 0$, which gives $kl = n\pi$, where $n \in \mathbb{Z}$. Since k must also be positive, we find its allowed values as

$$k = \frac{n\pi}{l}, \quad n = 1,\, 2,\, \ldots\ . \tag{2.11}$$

Substituting $v(x)$ and $q(t)$ from Eq. (2.8) into Eq. (2.3) gives the solution[12]

$$u(x,t) = C \sin \frac{n\pi x}{l} \cos(\omega_n t + \alpha), \tag{2.12}$$

where $\omega_n = n\pi c/l$. Such solutions of the wave equation are known as *standing waves*, and are also called *eigenvibrations*, or *normal modes*. Each of them vibrates at a specific frequency ω_n.

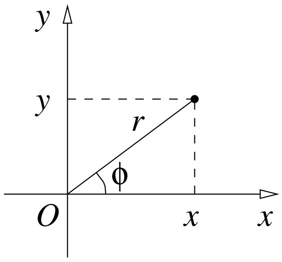
Note that if we took the separation constant λ to be positive, Eq. (2.9) would have the form $v'' - k^2 v = 0$. It solutions $v(x) = Ae^{kx} + Be^{-kx}$ cannot be made to satisfy $v(0) = v(l) = 0$ with nonzero A or B. Hence, $\lambda < 0$, as we assumed.

[10] Using the trigonometric addition formula, $\cos(\omega t + \alpha) = \cos \omega t \cos \alpha - \sin \omega t \sin \alpha$, we see that $C_1 = C \cos \alpha$ and $C_2 = -C \sin \alpha$, so the solutions (2.7) and (2.8) are equivalent.

[11] For a positive separation constant λ, the function $q(t)$ would behave exponentially, as $q(t) = C_1 e^{\sqrt{\lambda} t} + C_2 e^{-\sqrt{\lambda} t}$, which goes against our experience of the motion of strings. We can view this as a "physical proof" of the fact that $\lambda < 0$.

[12] We also denote the product of two arbitrary constants C and B by B.

2.3 Variable separation for a circular membrane. Bessel functions



Vibrations of a membrane are described by the two-dimensional wave equation (1.15). For a circular membrane, it is convenient to use plane polar coordinates r and ϕ for which $x = r\cos\phi$ and $y = r\sin\phi$. In this case, Eq. (1.15) takes the form[13]

$$\frac{\partial^2 u}{\partial t^2} - c^2\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \phi^2}\right] = 0. \tag{2.13}$$

We assume that the membrane is held fixed at the edge $r = a$ (like the skin or a drum), where a is the radius of the membrane. Hence, the solutions $u(r,\phi,t)$ of Eq. (2.13) must satisfy the boundary condition $u(a,\phi,t) = 0$.

Using variable separation, we seek solutions in the form

$$u(r,\phi,t) = v(r,\phi)q(t). \tag{2.14}$$

Substituting this into (2.13) and dividing through by $v(r,\phi)q(t)$ gives

$$\underbrace{\frac{1}{q(t)}\frac{d^2 q}{dt^2}}_{\lambda} - \underbrace{c^2\frac{1}{v(r,\phi)}\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 v}{\partial \phi^2}\right]}_{\lambda} = 0. \tag{2.15}$$

In this equation, the first term depends only on t, while the second one only on r and ϕ. Hence, for Eq. (2.15) to hold for all r, ϕ and t, each term must be equal to a constant λ. As in Sec. 2.2, we assume it to be negative, $\lambda = -\omega^2$.

For $q(t)$, we have the same equation as Eq. (2.6) with solution (2.8). The second term in Eq. (2.15) leads to the equation

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 v}{\partial \phi^2} + k^2 v = 0, \tag{2.16}$$

for the spatial part $v(r,\phi)$, where $k = \omega/c$. In the next step, we use

$$v(r,\phi) = R(r)\Phi(\phi), \tag{2.17}$$

to separate the radial and angular parts of the equation. Substituting this into Eq. (2.16), dividing it through by $R(r)\Phi(\phi)$ and rearranging gives

$$\frac{1}{R(r)}\frac{1}{r}\frac{d}{dr}\left(r\frac{dR}{dr}\right) + k^2 + \frac{1}{r^2}\underbrace{\frac{1}{\Phi(\phi)}\frac{d^2\Phi}{d\phi^2}}_{-m^2} = 0. \tag{2.18}$$

For this equation to hold for all r and ϕ, the factor that contains Φ and its second-order derivate must be equal to a constant. Assuming that this constant is non-positive[14] and denoting it by $-m^2$, yields the equation for the angular part:

$$\Phi'' + m^2\Phi = 0. \tag{2.19}$$

[13] We use the expression for the Laplacian in polar coordinates $\nabla^2 = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2}{\partial \phi^2}$.

[14] Only this choice allows $\Phi(\phi)$ to be 2π-periodic, as required.

Its general solution is

$$\Phi(\phi) = A\cos m\phi + B\sin m\phi, \tag{2.20}$$

and since $\Phi(\phi)$ is periodic with period 2π, we must have $m = 0,\ 1,\ 2, \ldots$.

Looking back at Eq. (2.18), we see that the radial part satisfies the equation[15]

$$\frac{d^2R}{dr^2} + \frac{1}{r}\frac{dR}{dr} + \left(k^2 - \frac{m^2}{r^2}\right)R = 0. \tag{2.21}$$

Dividing this equation by k^2,

$$\frac{d^2R}{d(kr)^2} + \frac{1}{kr}\frac{dR}{d(kr)} + \left(1 - \frac{m^2}{k^2r^2}\right)R = 0,$$

and introducing the new variable $z = kr$, gives the equation for $R(z)$,

$$\frac{d^2R}{dz^2} + \frac{1}{z}\frac{dR}{dz} + \left(1 - \frac{m^2}{z^2}\right)R = 0, \tag{2.22}$$

or

$$z^2R'' + zR' + (z^2 - m^2)R = 0. \tag{2.23}$$

Equations (2.22) and (2.23) are know as the *Bessel equations*[16]. Its solution can not be expressed through a combination of elementary functions (rational powers, exponential, logarithmic or trigonometric).

We thus need to introduce new functions that solve these equations. We seek them in the form of a power series, $R(z) = \sum_{n=0}^{\infty} a_n z^n$. Substituting $R(z)$, $R'(z) = \sum_{n=0}^{\infty} a_n n z^{n-1}$, $R''(z) = \sum_{n=0}^{\infty} a_n n(n-1) z^{n-2}$ into Eq. (2.23), gives

$$\sum_{n=0}^{\infty} a_n n(n-1)z^n + \sum_{n=0}^{\infty} a_n n z^n + \sum_{n=0}^{\infty} a_n z^{n+2} - m^2 \sum_{n=0}^{\infty} a_n z^n = 0. \tag{2.24}$$

Let us rename the summation index in the third sum by changing $n+2$ to n, so that $a_n z^{n+2}$ becomes $a_{n-2}z^n$, and setting $a_{-2} = a_{-1} = 0$. This allows one to write Eq. (2.24) as a single sum,

$$\sum_{n=0}^{\infty} \left[n(n-1)a_n + na_n + a_{n-2} - m^2 a_n\right] z^n = 0.$$

In order for this series to be zero identically, the expression in brackets must be zero, giving

$$(n^2 - m^2)a_n + a_{n-2} = 0,$$

which leads to the recurrence relation

$$a_n = -\frac{a_{n-2}}{(n-m)(n+m)}. \tag{2.25}$$

[15] This follows from $\dfrac{1}{r}\dfrac{d}{dr}\left(r\dfrac{dR}{dr}\right) = \dfrac{d^2R}{dr^2} + \dfrac{1}{r}\dfrac{dR}{dr}$.

[16] Friedrich Wilhelm Bessel (1784–1846), German astronomer, mathematician and physicist.

This relation allows one to generate the coefficients of the power series starting from some initial coefficient. Because of the $n-m$ factor in the denominator of Eq. (2.25), one must have $n > m$, so the first nonzero coefficient in the series is a_m ($m = 0, 1, 2, \ldots$). We also see from Eq. (2.25) that for an even m, all nonzero terms in the power series are even, while for an odd m all terms will be odd. Since Eq. (2.23) is homogeneous, the choice of a_m is a matter of convention, as the solution can always be multiplied by an arbitrary factor.

Let's start with the simplest case $m = 0$ for which Eq. (2.25) reads

$$a_n = -\frac{a_{n-2}}{n^2}.$$

Choosing $a_0 = 1$, we obtain

$$a_2 = -\frac{1}{2^2},\ a_4 = \frac{1}{2^2 \cdot 4^2},\ a_6 = -\frac{1}{2^2 \cdot 4^2 \cdot 6^2},\ \ldots,\ a_{2k} = \frac{(-1)^k}{2^{2k}(k!)^2},\ \ldots,$$

which gives the solution in the form

$$R(z) = 1 - \frac{z^2}{2^2} + \frac{z^4}{2^4(2!)^2} - \frac{z^6}{2^6(3!)^2} + \ldots = \sum_{k=0}^{\infty}(-1)^k \frac{(z/2)^{2k}}{(k!)^2}.$$

This function is the *Bessel function of the first kind* of order zero, denoted by

$$J_0(z) = \sum_{k=0}^{\infty}(-1)^k \frac{(z/2)^{2k}}{(k!)^2}. \tag{2.26}$$

Similarly, for $m > 0$, sataring from $a_m \neq 0$, we have from Eq. (2.25),

$$\begin{aligned}
a_{m+2} &= -\frac{a_m}{(m+2-m)(m+m+2)} = -\frac{a_m}{2(2m+2)}, \\
a_{m+4} &= \frac{a_{m+2}}{(m+4-m)(m+m+4)} = \frac{a_m}{2 \cdot 4(2m+2)(2m+4)}, \\
a_{m+6} &= -\frac{a_{m+4}}{(m+6-m)(m+m+6)} = -\frac{a_m}{2 \cdot 4 \cdot 6(2m+2)(2m+4)(2m+6)},
\end{aligned}$$

etc., so that

$$\begin{aligned}
a_{m+2} &= -\frac{a_m}{2^2(m+1)}, \\
a_{m+4} &= \frac{a_m}{2^4 2!(m+1)(m+2)}, \\
a_{m+6} &= -\frac{a_m}{2^6 3!(m+1)(m+2)(m+3)}, \\
a_{m+2k} &= (-1)^k \frac{a_m}{2^{2k} k!(m+1)\ldots(m+k)}.
\end{aligned}$$

It is convenient to choose $a_m = 1/(2^m m!)$, so that $R(z) = J_m(z)$, where

$$J_m(z) = \left(\frac{z}{2}\right)^m \sum_{k=0}^{\infty}(-1)^k \frac{(z/2)^{2k}}{k!(m+k)!} \tag{2.27}$$

is the Bessel function of the first kind of order m.

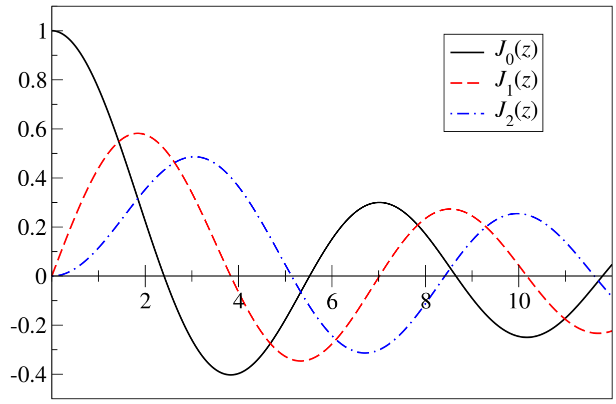


Figure 2: Bessel functions $J_m(z)$ for $m = 0, 1, 2$.

At small z ($z \ll 1$), $J_m(z) \simeq (z/2)^m/m!$, so that $J_0(0) = 1$ and $J_m(0) = 0$ for $m > 0$. At larger z the Bessel functions oscillate, somewhat similarly to the cosine and sine functions, but with a decreasing amplitude, see Fig. 2.

Each of the Bessel functions has infinitely many zeros. The nth zero of $J_m(z)$ is denoted $z_{m,n}$ ($n = 1, 2, \dots$). Approximate values of the zeros for small m and n are given in the table 1.[17]

Table 1: Zeros $z_{m,n}$ of the functions $J_m(z)$ for $m = 0, \dots, 4$ and $n = 1, \dots, 5$.

$m\backslash n$	1	2	3	4	5
0	2.4048	5.5201	8.6537	11.7915	14.9309
1	3.8317	7.0156	10.1735	13.3237	16.4706
2	5.1356	8.4172	11.6198	14.7960	17.9598
3	6.3802	9.7610	13.0152	16.2235	19.4094
4	7.5883	11.0647	14.3725	17.616	20.8269

Thus the solution of the radial equation (2.21) which is finite at $r = 0$ is $R(r) = J_m(kr)$.[18] To satisfy the boundary condition $u(a, \phi, t) = 0$, we require $R(a) = 0$, i.e., $J_m(ka) = 0$. This means that $ka = z_{m,n}$, which gives

$$k = \frac{z_{m,n}}{a}. \tag{2.28}$$

Hence the frequencies of harmonic vibrations of the membrane are determined by the zeros of the Bessel functions,

$$\omega = kc = \frac{z_{m,n}}{a} c \equiv \omega_{m,n}. \tag{2.29}$$

The corresponding displacement $u(r, \phi, t)$ of the membrane is obtained by combining the radial, angular and temporal parts of the solution,

$$u(r, \varphi, t) = J_m\left(\frac{z_{m,n}}{a} r\right) [A \cos m\varphi + B \sin m\varphi] \cos(\omega_{m,n} t + \alpha), \tag{2.30}$$

[17] For $n \gg m$ the roots are given by $z_{m,n} \approx \left(n + \frac{m}{2} - \frac{1}{4}\right)\pi$.

[18] Any second-order differential equation has two linearly independent solutions. However, the second solution of the Bessel equation has a singularity (i.e., diverges) at $z = 0$. These solutions are usually taken in the form of the Bessel functions of the second kind, denoted $Y_n(z)$ or $N_n(z)$, and sometimes called the Weber or Neumann functions.

Note that one can always choose the $\varphi = 0$ direction so that $B = 0$, and the angular dependence is given by $\cos m\varphi$.

The shape of the vibrating membrane can be visualised by showing positive $(+)$ and negative $(-)$ displacements at a given time instant (e.g., $t = 0$ and for $\alpha = 0$), see Fig. 3. The areas of positive and negative displacements are separated by *nodal lines* shown by dashed lines.[19]

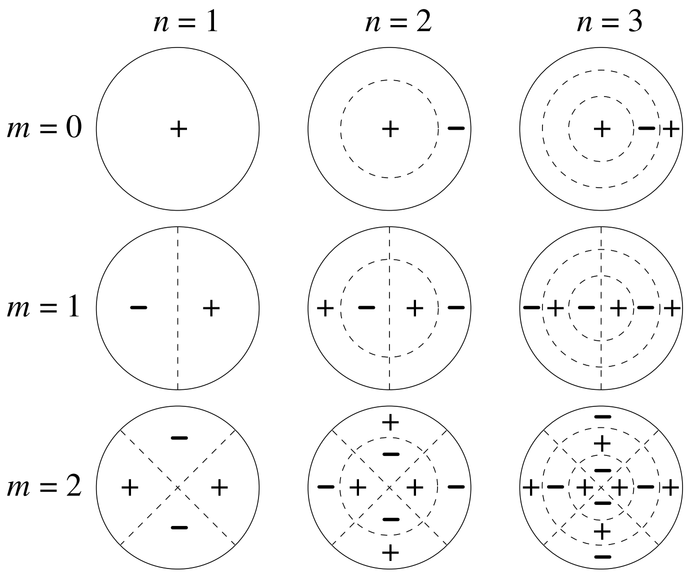


Figure 3: Shapes of harmonic vibrations (*normal modes*) of the circular membrane for $m = 0,\ 1,\ 2$ and $n = 1,\ 2,\ 3$.

2.4 Variable separation for the heat equation

Consider the heat equation in one dimension for $u(x, t)$, Eq. (1.27),

$$\frac{\partial u}{\partial t} - K\frac{\partial^2 u}{\partial x^2} = 0, \tag{2.31}$$

for a rod of length l $(0 \le x \le l)$, whose ends are kept at zero temperature,

$$u(0, t) = 0, \quad u(l, t) = 0. \tag{2.32}$$

Seeking solution in the form

$$u(x, t) = v(x)q(t), \tag{2.33}$$

we substitute (2.33) into (2.31) and obtain:

$$v(x)\frac{dq}{\partial t} - q(t)K\frac{d^2 v}{dx^2} = 0. \tag{2.34}$$

Dividing (2.34) through by $v(x)q(t)$ yields

$$\underbrace{\frac{1}{q(t)}\frac{dq}{dt}}_{\text{const}} - \underbrace{\frac{K}{v(x)}\frac{d^2 v}{dx^2}}_{\text{const}} = 0. \tag{2.35}$$

[19] A drum or tympani player knows that the sound the instrument makes depends on where the skin is struck. This is because by striking close or further away from the centre one produces more or less contribution of the different types of vibrations shown in Fig. 3.

In this equation the first term depends only on t, while the second one only on x. For the equation to be valid for all x and t, the two terms must be equal to a constant. Denoting the separation constant by $-\lambda$ (with $\lambda > 0$ as we shall see below), we have from the first term in (2.35):

$$\frac{1}{q(t)}\frac{dq}{dt} = -\lambda,$$

or

$$\frac{dq}{dt} = -\lambda q(t). \tag{2.36}$$

Solving this equation,

$$\int \frac{dq}{q} = -\int \lambda dt,$$
$$\ln q = -\lambda t + C,$$
$$q = e^C e^{-\lambda t},$$

and replacing e^C by C, where C is an arbitrary constant, we obtain

$$q(t) = Ce^{-\lambda t}. \tag{2.37}$$

Note that for $\lambda > 0$ this is a decreasing function of time, while for $\lambda < 0$, $q(t) \to \infty$ for $t \to \infty$. Hence, adopting $\lambda < 0$ would lead to an unrealistic heating of the rod. This is a physical justification of our choice of the sign of the separation constant.

The coordinate part of (2.35) reads

$$\frac{K}{v(x)}\frac{d^2v}{dx^2} = -\lambda,$$

or

$$\frac{d^2v}{dx^2} + \frac{\lambda}{K}v = 0. \tag{2.38}$$

This equation has the general solution,

$$v(x) = A\cos\left(\sqrt{\frac{\lambda}{K}}\,x\right) + B\sin\left(\sqrt{\frac{\lambda}{K}}\,x\right), \tag{2.39}$$

where A and B are arbitrary constants. To satisfy the boundary conditions (2.32) for $u(x,t)$ in the form (2.33), we require

$$v(0) = 0, \quad v(l) = 0. \tag{2.40}$$

Applying the first of these to (2.39) gives $A = 0$, and the second then reads

$$B\sin\left(\sqrt{\frac{\lambda}{K}}\,l\right) = 0.$$

As $B \neq 0$ for a nonzero $v(x)$, the argument of the sine function must satisfy

$$\sqrt{\lambda/K}\,l = n\pi, \qquad n = 1, 2, \ldots .$$

This determines the possible values of the separation constant[20]

$$\lambda = \frac{n^2\pi^2}{l^2} K, \tag{2.41}$$

and the corresponding solutions

$$v(x) = B \sin \frac{n\pi x}{l}. \tag{2.42}$$

Combining Eqs. (2.37) and (2.42), and using (2.41), we obtain

$$u(x,t) = B_n \sin \frac{n\pi x}{l}\, e^{-(n^2\pi^2/l^2)Kt}, \tag{2.43}$$

where B_n is a new arbitrary constant, and $n = 1, 2, \ldots$.

By the *superposition principle*, one can obtain a more general solution of equation (2.31) with boundary conditions (2.32), as a superposition (i.e., linear combination) of solutions (2.43) with different n:

$$u(x,t) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{l}\, e^{-(n^2\pi^2/l^2)Kt}$$

The coefficients B_n can be determined from the initial condition, e.g.,

$$u(x,0) = f(x), \tag{2.44}$$

where $f(x)$ describes the temperature distribution in the rod at $t = 0$. This gives

$$\sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{l} = f(x). \tag{2.45}$$

To find B_n, we can multiply (2.45) by $\sin(m\pi x/l)$ and integrate it from 0 to l using the identity[21]

$$\int_0^l \sin \frac{n\pi x}{l} \sin \frac{m\pi x}{l}\, dx = \frac{l}{2}\delta_{nm}, \quad m, n = 1, 2, \ldots, \tag{2.46}$$

where δ_{mn} is the Kronecker delta symbol:

$$\delta_{mn} = \begin{cases} 1, & m = n, \\ 0, & m \neq n. \end{cases} \tag{2.47}$$

After such integration, the only nonzero term in the sum is that with $n = m$, which yields

$$B_m = \frac{2}{l} \int_0^l f(x) \sin \frac{m\pi x}{l}\, dx. \tag{2.48}$$

The series on the left-hand side of (2.45) is a *Fourier series*. We study Fourier series in Chapter 3. In particular, one needs to verify that substitution of (2.48) into (2.45) gives an identity for any well-behaved function $f(x)$.

[20] For $\lambda < 0$, the general solution of (2.38) is $v(x) = Ae^{\sqrt{-\lambda/K}x} + Be^{-\sqrt{-\lambda/K}x}$, and the boundary conditions (2.40) cannot be satisfied for nonzero A or B.

[21] To verify (2.46), one can use $\sin\alpha \sin\beta = \frac{1}{2}[\cos(\alpha-\beta) - \cos(\alpha+\beta)]$.

3 Fourier series

As we saw in Ch. 2, solutions of common linear homogeneous PDEs with homogeneous boundary conditions contain sine or cosine functions. Using the superposition principle (Sec. 1.6), these solutions can be combined into more general solutions which take the form of infinite sums with some coefficients. Such sums are know as Fourier series.

The question we would like to answer is whether these solutions represent the general solution of the PDE with given boundary conditions, and whether one can choose the coefficients in such a way the the solution would satisfy particular initial conditions, e.g., $u(x,0) = f(x)$ for the heat equation in 1D, or $u(x,0) = f(x)$ and $u_t(x,0) = g(x)$ for the wave equation for a string.

3.1 Fourier expansion and convergence

We know that if a function $f(x)$ is differentiable infinitely many times at a point, e.g., $x = 0$, then it can be represented by an expansion in powers of x, i.e., $x^0 = 1$, x, x^2, $\ldots$,

$$f(x) = \sum_{n=0}^{\infty} a_n x^n, \quad \text{where} \quad a_n = \frac{f^{(n)}(0)}{n!}, \tag{3.1}$$

called the Taylor series (or Maclaurin series, for expansions about $x = 0$). If the series converges and the function is infinitely differentiable on an open interval that contains $x = 0$, then the series converges to $f(x)$, as implied by the equal sign in the first of Eq. (3.1).

Let us now consider a "well-behaved" function $f(x)$ on the interval $(-\pi, \pi)$. This function can be represented by its *Fourier series*,

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx), \tag{3.2}$$

where the coefficients are given by

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx, \tag{3.3}$$

and where $f(x)$ will actually be equal to the series (3.2) at almost all points on $(-\pi, \pi)$.

The expansion (3.2) with the relations (3.3) hold because the trigonometric functions

$$1, \quad \cos x, \quad \sin x, \quad \cos 2x, \quad \sin 2x, \ldots \tag{3.4}$$

form a complete orthogonal system of functions on $(-\pi, \pi)$, i.e., serve as a *basis*. The coefficients a_n and b_n are then "components" of $f(x)$ in this basis.

The Fourier series for a function is similar to the basis expansion of a vector, e.g., $\mathbf{a} = a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k}$, where $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are unit vectors along the axes x, y and z, orthogonal to each other ($|\mathbf{i}| = |\mathbf{j}| = |\mathbf{k}| = 1$, $\mathbf{i}\cdot\mathbf{j} = 0$, etc.). Equations (3.3) are analogs of the relations $a_x = \mathbf{a}\cdot\mathbf{i}$, $a_y = \mathbf{a}\cdot\mathbf{j}$, $a_z = \mathbf{a}\cdot\mathbf{k}$, , with the integrals in Eq. (3.3) playing the role of the scalar product.

Indeed, the functions (3.4) obey the following *orthogonality relations*,

$$\int_{-\pi}^{\pi} \cos mx \cos nx\, dx = \begin{cases} 2\pi, & m = n = 0, \\ \pi, & m = n > 0, \\ 0, & m \neq n, \end{cases} \tag{3.5}$$

$$\int_{-\pi}^{\pi} \sin mx \sin nx\, dx = \begin{cases} \pi, & m = n, \\ 0, & m \neq n, \end{cases} \tag{3.6}$$

$$\int_{-\pi}^{\pi} \sin mx \cos nx\, dx = 0, \tag{3.7}$$

where $m, n \in \mathbb{Z}$.

Let us verify these identities. For the case of $m = n = 0$ in Eq. (3.5), we have:

$$\int_{-\pi}^{\pi} \cos 0 \cos 0\, dx = \int_{-\pi}^{\pi} dx = [x]_{-\pi}^{\pi} = \pi - (-\pi) = 2\pi.$$

For $m = n > 0$, we use the double-angle formula $\cos 2\theta = 2\cos^2\theta - 1$ in the form $\cos^2\theta = \frac{1}{2}(1 + \cos 2\theta$, to integrate:

$$\begin{aligned} \int_{-\pi}^{\pi} \cos^2 nx\, dx &= \frac{1}{2}\int_{-\pi}^{\pi} (1 + \cos 2nx) dx \\ &= \frac{1}{2}\int_{-\pi}^{\pi} dx + \frac{1}{2}\int_{-\pi}^{\pi} \cos 2nx\, dx \\ &= \pi + \frac{1}{4n}[\sin 2nx]_{-\pi}^{\pi} \\ &= \pi + \frac{1}{4n}\big[\underbrace{\sin(2n\pi)}_{=0} - \underbrace{\sin(-2n\pi)}_{=0}\big] \\ &= \pi, \end{aligned}$$

as in Eq. (3.5), second line.

Finally, for $m \neq n$, we use the identity $\cos\alpha\cos\beta = \frac{1}{2}[\cos(\alpha+\beta) + \cos(\alpha-\beta)]$:

$$\begin{aligned} \int_{-\pi}^{\pi} \cos mx \cos nx\, dx &= \frac{1}{2}\int_{-\pi}^{\pi} \big[\cos(m+n)x + \cos(m-n)x\big] dx \\ &= \frac{1}{2}\left\{\frac{1}{m+n}[\sin(m+n)x]_{-\pi}^{\pi} + \frac{1}{m-n}[\sin(m-n)x]_{-\pi}^{\pi}\right\} \\ &= 0, \end{aligned}$$

since $\sin\nu\pi = 0$ for $\nu \in \mathbb{Z}$. This proves the last line of Eq. (3.5).

Equations (3.6) and (3.7) are verified similarly using the double-angle formula $\cos 2\theta = 1 - 2\sin^2\theta$, as well as $\sin\alpha\sin\beta = \frac{1}{2}[\cos(\alpha-\beta) - \cos(\alpha+\beta)]$ and $\sin\alpha\cos\beta = \frac{1}{2}[\sin(\alpha+\beta) + \sin(\alpha-\beta)]$.[22]

[22] It is worthwhile to perform the relevant calculations.

To check that the series (3.2) with the coefficients from Eq. (3.3) represents the function $f(x)$, let us assume the series converges to the values of $f(x)$, i.e., that

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx). \tag{3.8}$$

Multiplying this equation by $\cos mx$ with $m > 0$ and integrating between $-\pi$ and π, we have

$$\int_{-\pi}^{\pi} f(x)\cos mx\, dx = \frac{a_0}{2}\int_{-\pi}^{\pi}\cos mx\, dx + \sum_{n=1}^{\infty} a_n \int_{-\pi}^{\pi}\cos nx \cos mx\, dx$$
$$+ \sum_{n=1}^{\infty} b_n \int_{-\pi}^{\pi}\sin nx \cos mx\, dx.$$

In the first integral on the right-hand side, the integrand can be written as $\cos mx = \cos 0x \cos mx$ (since $\cos 0x = 1$). From Eq. (3.5), such integral is equal to zero for $m \neq 0$. The integral in the first sum on the right-hand side is equal to π for $n = m$, but is zero otherwise, see Eq. (3.5). This means that the only nonzero term in the sum over n is that with $n = m$, and it is equal to $a_m\pi$. The integral in the second sum is given by Eq. (3.7) with n and m interchanged, and is equal to zero. Hence, we have

$$\int_{-\pi}^{\pi} f(x)\cos mx\, dx = \pi a_m,$$

or

$$a_m = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos mx\, dx,$$

which is the same as the first expression in Eq. (3.3). Similarly, multiplying Eq. (3.8) by $\sin mx$ and integrating between $-\pi$ and π, we find

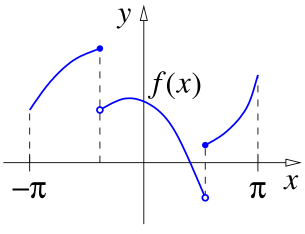
$$\int_{-\pi}^{\pi} f(x)\sin mx\, dx = \pi b_m,$$

leading to the expression for b_n in Eq. (3.3). Finally, integrating Eq. (3.8) between $-\pi$ and π gives

$$\int_{-\pi}^{\pi} f(x)dx = \frac{a_0}{2}\int_{-\pi}^{\pi} dx \quad \Rightarrow \quad a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx,$$

since $\int_{-\pi}^{\pi}\cos nx dx = \int_{-\pi}^{\pi}\sin nx dx = 0$ for $n > 0$, and in accord with Eq. (3.3).

In Eq. (3.8), we assumed that the Fourier series converges to $f(x)$ for all values of x. This may not hold in general, but is true for "well-behaved" functions, as described in the following theorem.



Convergence Theorem. Let f be a piecewise smooth function[23] on $(-\pi, \pi)$, and let all its discontinuities be finite "jumps". Then the Fourier series (3.2) converges to $f(x)$ for all x where f is continuous. If f has a discontinuity at $x = \xi$, then the Fourier series converges to $\frac{1}{2}[f(\xi - 0) + f(\xi + 0)]$.[24]

[23] A piecewise smooth function is a function whose derivative is piecewise continuous.
[24] $f(\xi - 0) = \lim_{x\to\xi-0} f(x)$ and $f(\xi + 0) = \lim_{x\to\xi+0} f(x)$ are the left-hand and right-hand limits of $f(x)$ at $x = \xi$.

The Fourier series is periodic with period 2π. It provides a *periodic extension* of $f(x)$ onto the whole real axis. At $x = -\pi$ and $x = \pi$ the Fourier series equals $\frac{1}{2}[f(\pi - 0) + f(-\pi + 0)]$. Figure 4 shows the function $f(x) = x$, $x \in (-\pi, \pi)$, and its periodic extension[25]. The periodic extension has discontinuities at $\pm\pi$, $\pm 3\pi$, etc., at which its values are $\frac{1}{2}[f(\pi - 0) + f(-\pi + 0)] = \frac{1}{2}[\pi + (-\pi)] = 0$.

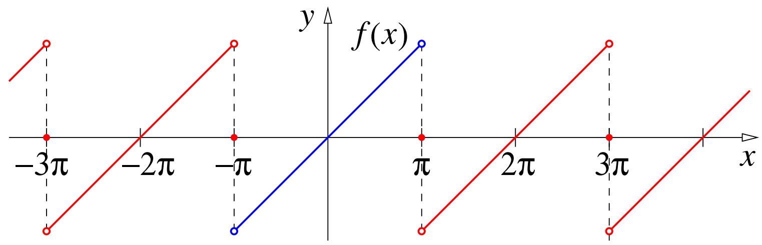


Figure 4: The function $f(x) = x$, $x \in (-\pi, \pi)$ (blue) and its periodic extension given by the Fourier series (red).

3.2 Fourier series for even and odd functions

The Fourier series (3.2) looks simpler when the function $f(x)$ is even or odd[26].

For an *even* function $f(x)$, the product $f(x) \sin nx$ is an odd function. Its integral over any symmetric interval is zero. Equation (3.3) then gives

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx = 0.$$

Hence, the Fourier series for an even function has the form

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx. \tag{3.9}$$

The expression for the coefficients a_n (3.3) can also be simplified. The product $f(x) \cos nx$ is an even function. Therefore,

$$\int_{-\pi}^{0} f(x) \cos nx \, dx = \int_{0}^{\pi} f(x) \cos nx \, dx,$$

and we have

$$a_n = \frac{2}{\pi} \int_{0}^{\pi} f(x) \cos nx \, dx. \tag{3.10}$$

For an *odd* function $f(x)$, the product $f(x) \cos nx$ is an odd function, and

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx = 0.$$

The Fourier series for an odd function then has the form

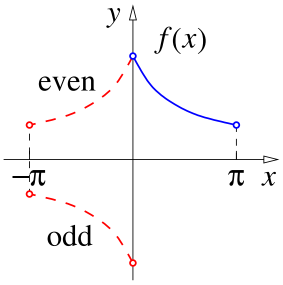
$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx, \tag{3.11}$$

[25] It is not hard to show that the Fourier series for this function is $2 \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \sin nx$.

[26] A function $f(x)$ with a symmetric domain (i.e., that which contains both x in $-x$) is even if $f(-x) = f(x)$ and is odd if $f(-x) = -f(x)$.

and since $f(x)\sin nx$ is an even function, we have

$$b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\,dx. \tag{3.12}$$



Half-range Fourier series.

The coefficients in Eq. (3.10) or (3.12) are determined by the values of $f(x)$ on the interval $(0,\pi)$. Hence, we can use the Fourier expansion in the form of either Eq. (3.9) or Eq. (3.11) for a function defined on $0 < x < \pi$.

Expanding $f(x)$ in the cosine Fourier series (3.9) provides an even extension of $f(x)$, while using the sine Fourier series (3.11) gives an odd extension of this function. The cosine Fourier series converges to $f(0)$ and $f(\pi)$ at $x = 0$ and $x = \pi$, respectively (assuming $f(x)$ is defined for $x = 0$ and π). By contrast, the sine Fourier series (3.11) will always have a value of zero at $x = 0$ and π.

3.3 Complex form of the Fourier series

Let us use Euler's formula and its complex conjugate,

$$e^{i\theta} = \cos\theta + i\sin\theta, \quad e^{-i\theta} = \cos\theta - i\sin\theta, \tag{3.13}$$

to write the Fourier series (3.8) in the complex form. Adding and subtracting Eqs. (3.13), we find

$$\cos\theta = \frac{1}{2}\left(e^{i\theta} + e^{-i\theta}\right), \quad \sin\theta = \frac{1}{2i}\left(e^{i\theta} - e^{-i\theta}\right).$$

Using these to express $\cos nx$ and $\sin nx$ in terms of e^{inx} and e^{-inx}, we have

$$\begin{aligned} f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n\cos nx + b_n\sin nx) \\ &= \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n\frac{1}{2}\left(e^{inx} + e^{-inx}\right) + b_n\frac{1}{2i}\left(e^{inx} - e^{-inx}\right)\right] \\ &= \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[\left(\frac{a_n}{2} + \frac{b_n}{2i}\right)e^{inx} + \left(\frac{a_n}{2} - \frac{b_n}{2i}\right)e^{-inx}\right] \end{aligned}$$

Introducing

$$c_n = \frac{a_n}{2} + \frac{b_n}{2i} = \frac{a_n - ib_n}{2} \quad \text{and} \quad c_{-n} = \frac{a_n}{2} - \frac{b_n}{2i} = \frac{a_n + ib_n}{2}, \tag{3.14}$$

with $c_0 = a_0/2$ [consistent with $b_0 = 0$, as $\sin 0x = 0$ in Eq. (3.3)], we have

$$f(x) = c_0 + \sum_{n=1}^{\infty} c_n e^{inx} + \sum_{n=1}^{\infty} c_{-n}e^{-inx}.$$

This can be written as a single sum over integer n from $-\infty$ to ∞,

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}. \tag{3.15}$$

The coefficients c_n defined in Eq. (3.14) can be found using Eq. (3.3),

$$\begin{aligned} c_n &= \frac{1}{2}a_n + \frac{1}{2i}b_n \\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx - \frac{i}{2\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx \\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)(\cos nx - i\sin nx)dx, \end{aligned}$$

which gives

$$c_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-inx} f(x)dx. \tag{3.16}$$

The complex form of the Fourier expansion described by Eqs. (3.15) and (3.16) is very compact and neat [cf. Eqs. (3.2) and (3.3)]. In fact, it is easy to obtain Eq. (3.16) directly from Eq. (3.15), using the orthogonality relation[27]

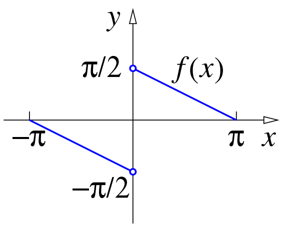
$$\int_{-\pi}^{\pi} e^{-imx}e^{inx}dx = 2\pi\delta_{mn}. \tag{3.17}$$

Multiplying (3.15) by e^{-imx} and integrating from $-\pi$ to π, we have

$$\int_{-\pi}^{\pi} e^{-imx} f(x)dx = \sum_{n=-\infty}^{\infty} c_n \int_{-\pi}^{\pi} e^{-imx}e^{inx}dx = \sum_{n=-\infty}^{\infty} c_n 2\pi\delta_{mn} = 2\pi c_m,$$

since only the term with $n = m$ contributes to the sum, because of the Kronecker delta. The last equation gives the expression for c_m which is the same as that in Eq. (3.16).

3.4 Examples of Fourier series



Example 1. Let us derive the Fourier expansion for the function

$$f(x) = \begin{cases} \dfrac{\pi - x}{2}, & 0 < x \le \pi, \\ -\dfrac{\pi + x}{2}, & -\pi \le x < 0. \end{cases} \tag{3.18}$$

This is an odd function. Hence, $a_n = 0$ (see Sec. 3.2), and

$$\begin{aligned} b_n &= \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\,dx \\ &= \frac{2}{\pi}\cdot\frac{1}{2}\int_0^{\pi}(\pi - x)\sin nx\,dx \\ &= \frac{1}{\pi}\left[\pi\int_0^{\pi}\sin nx\,dx - \int_0^{\pi} x\sin nx\,dx\right]. \end{aligned}$$

[27] Indeed, for $m, n \in \mathbb{Z}$ and $m \ne n$, we have

$$\int_{-\pi}^{\pi} e^{-imx}e^{inx}dx = \frac{1}{i(n-m)}\left[e^{i(n-m)x}\right]_{-\pi}^{\pi} = \frac{1}{i(n-m)}\left[e^{i(n-m)\pi} - e^{-i(n-m)\pi}\right] = 0,$$

since $e^{i\pi} = e^{-i\pi} = -1$, while for $m = n$, $\int_{-\pi}^{\pi} e^{-imx}e^{inx}dx = \int_{-\pi}^{\pi} dx = 2\pi$.

Using integration by parts in the second integral, we obtain

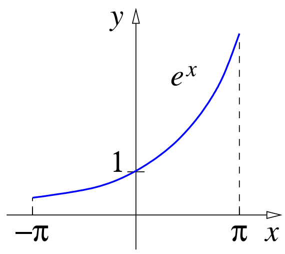
$$
\begin{aligned}
b_n &= \left[-\frac{\cos nx}{n}\right]_0^\pi + \frac{1}{\pi}\left[x\frac{\cos nx}{n}\right]_0^\pi - \frac{1}{\pi}\int_0^\pi \cos nx\,dx \\
&= -\frac{\cos n\pi}{n} + \frac{1}{n} + \frac{1}{\pi}\,\pi\frac{\cos n\pi}{n} - \frac{1}{n^2}[\sin nx]_0^\pi \\
&= \frac{1}{n},
\end{aligned}
$$

since $\sin n\pi = 0$. Hence, the Fourier expansion for $f(x)$ is

$$
f(x) = \sum_{n=1}^{\infty} \frac{\sin nx}{n}, \tag{3.19}
$$

which is probably the simplest Fourier expansion one encounters!

Note that for $x = 0$ the original function $f(x)$, Eq. (3.18), is undefined, but the Fourier series gives the value of 0. This is in agreement with the convergence theorem (Sec. 3.1) which predicts $\frac{1}{2}[f(0+0) + f(0-0)] = \frac{1}{2}[\pi/2 - \pi/2] = 0$.



Example 2. Let us now consider the function $f(x) = e^x$ on $-\pi < x < \pi$. This function is neither even nor odd, so we need to use the integrals from Eq. (3.3) to find the coefficients a_n and b_n. However, it is instructive and easier to derive the Fourier series in the complex form (3.15) using Eq. (3.16). We then have

$$
\begin{aligned}
c_n &= \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-inx}e^x dx \\
&= \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{(1-in)x} dx \\
&= \frac{1}{2\pi}\frac{1}{1-in}\left[e^{(1-in)x}\right]_{-\pi}^{\pi} \\
&= \frac{1}{2\pi}\frac{e^{(1-in)\pi} - e^{-(1-in)\pi}}{1-in} \\
&= \frac{1}{2\pi}\frac{e^{\pi}e^{-in\pi} - e^{-\pi}e^{in\pi}}{1-in} \\
&= \frac{1}{2\pi}\frac{(-1)^n\left(e^{\pi} - e^{-\pi}\right)}{1-in} \\
&= \frac{1}{\pi}\frac{(-1)^n \sinh\pi}{1-in},
\end{aligned} \tag{3.20}
$$

where we used $e^{in\pi} = \left(e^{i\pi}\right)^n = (-1)^n$ and $e^{-in\pi} = \left(e^{-i\pi}\right)^n = (-1)^n$, and the definition of the hyperbolic functions[28]. Hence, the Fourier expansion for $-\pi < x < \pi$ is

$$
e^x = \frac{\sinh\pi}{\pi}\sum_{n=-\infty}^{\infty}\frac{(-1)^n}{1-in}e^{inx}.
$$

To find the original, real form of the Fourier series (3.2) (i.e., in terms of $\cos nx$ and $\sin nx$), we can use the relation (3.14) between the coefficients a_n, b_n and c_n, which gives

$$
a_n = 2\,\mathrm{Re}\, c_n, \qquad b_n = -2\,\mathrm{Im}\, c_n.
$$

[28] The hyperbolic cosine and sine are $\cosh x = \frac{1}{2}\left(e^x + e^{-x}\right)$ and $\sinh x = \frac{1}{2}\left(e^x - e^{-x}\right)$.

Manipulating the expression for c_n, Eq. (3.20), we have

$$c_n = \frac{1}{\pi}\frac{(-1)^n \sinh\pi}{1-in} = \frac{1}{\pi}\frac{(-1)^n \sinh\pi(1+in)}{(1-in)(1+in)} = \frac{1}{\pi}\frac{(-1)^n \sinh\pi(1+in)}{1+n^2},$$

so that

$$a_n = \frac{2}{\pi}\frac{(-1)^n \sinh\pi}{1+n^2}, \qquad b_n = -\frac{2}{\pi}\frac{(-1)^n n\sinh\pi}{1+n^2}.$$

Using this in Eq. (3.2) gives the following Fourier expansion

$$e^x = \frac{\sinh\pi}{\pi} + \frac{2\sinh\pi}{\pi}\sum_{n=1}^{\infty}(-1)^n\left(\frac{1}{1+n^2}\cos nx - \frac{n}{1+n^2}\sin nx\right). \qquad (3.21)$$

This equation is valid for $-\pi < x < \pi$. The Fourier series on the right-hand side provides a periodic extension for e^x onto the whole real axis, see Fig. 5. At $x = \pm\pi$, $\pm 3\pi$, etc., the series converges to $\frac{1}{2}\left(e^{\pi} + e^{-\pi}\right) = \cosh\pi$.

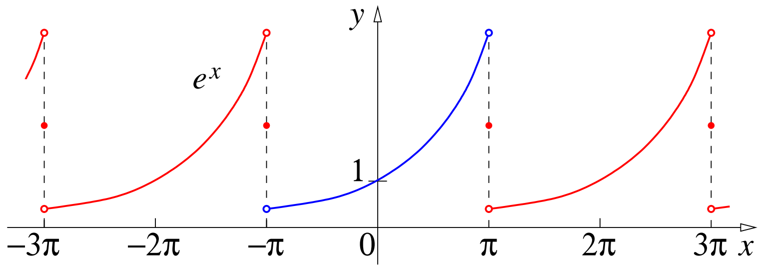


Figure 5: The function $f(x) = e^x$, $x \in (-\pi, \pi)$ (blue) and its periodic extension given by the Fourier series in Eq. (3.21) (red).

3.5 Fourier series on an arbitrary symmetric interval

Up to know we considered Fourier series on the interval from $-\pi$ to π. It is easy to generalise these results to an arbitrary symmetric interval by simply rescaling the independent variable x.

Let $f(x)$ be defined on $[-l, l]$. The Fourier series for this function can be obtained from (3.2) and (3.3) by a variable change, $x \to \pi x/l$, which gives

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n\cos\frac{n\pi x}{l} + b_n\sin\frac{n\pi x}{l}\right). \qquad (3.22)$$

For the expansion coefficients, we also change $dx \to (\pi/l)dx$, and obtain

$$a_n = \frac{1}{l}\int_{-l}^{l} f(x)\cos\frac{n\pi x}{l}\,dx, \qquad b_n = \frac{1}{l}\int_{-l}^{l} f(x)\sin\frac{n\pi x}{l}\,dx. \qquad (3.23)$$

As in Sec. 3.2, one can obtain a half-range cosine Fourier series for $f(x)$ on $[0, l]$, as an "even extension" of $f(x)$,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n\cos\frac{n\pi x}{l}, \qquad (3.24)$$

with

$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}\,dx. \tag{3.25}$$

Similarly, one can develop a half-range sine Fourier series (an "odd extension"),

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{l}, \tag{3.26}$$

with

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx. \tag{3.27}$$

These types of series will apply when we use the Fourier method to solve the wave equation or heat equation in one dimension on the interval $[0, l]$.

3.6 Parseval's identity and Gibbs phenomenon

This section is supplementary reading. Parceval's identity is an equation that links the sum of the squares of the Fourier coefficients for a given function and the integral of the square of this function. Gibbs phenomenon is a convergence pattern displayed by the Fourier series at the points where the function is discontinuous. It shows that no matter how many terms are included in the Fourier sum, the latter always overshoots or undershoots the one-sided limits of the function by a fixed amount (about 9% of the difference between the two one-sided limits).

Parseval's identity. Consider a piecewise smooth function $f(x)$ on $(-\pi, \pi)$ with a convergent Fourier series (3.8). Squaring both sides of this equation and integrating between $-\pi$ and π, we have

$$\begin{aligned}\int_{-\pi}^{\pi} f^2(x)dx = \int_{-\pi}^{\pi} &\left[\frac{a_0}{2} + \sum_{m=1}^{\infty}(a_m\cos mx + b_m\sin mx)\right] \\ \times &\left[\frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n\cos nx + b_n\sin nx)\right] dx.\end{aligned}$$

Here we have written the square on the right-hand side as a product of two identical Fourier sums with different summation indices, m and n, to avoid confusion.

Expanding the product of the two square brackets on the right-hand side and changing the order of summation and integration,[29] we obtain

$$\begin{aligned}\int_{-\pi}^{\pi}\left(\frac{a_0}{2}\right)^2 dx + \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} &\left[\int_{-\pi}^{\pi} a_m a_n \cos mx\cos nx\,dx\right. \\ &\left. + \int_{-\pi}^{\pi} b_m b_n \sin mx \sin nx\,dx\right] + \dots\end{aligned} \tag{3.28}$$

where the dots $(\dots)$ stands for all the other contributions, such as $\frac{a_0}{2}\int_{-\pi}^{\pi} a_n\cos nx\,dx$, $\int_{-\pi}^{\pi} a_m b_n \cos mx \sin nx\,dx$, etc. Due to orthogonality relations (3.5)–(3.7), these contributions vanish, and only the first term and those with $m = n$ in the double sum

[29] This requires uniform convergence of the Fourier series, which is guaranteed for a piecewise smooth function on every closed interval on which the function is continuous, R. Courant and D. Hilbert, *Methods of Mathematical Physics*, vol. 1 (Interscience Publishers, New York, 1953) Ch. II, § 5.

in (3.28) are nonzero. Hence,

$$\int_{-\pi}^{\pi} f^2(x)dx = \pi\left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty}(a_n^2 + b_n^2)\right], \tag{3.29}$$

which is known as *Parseval's identity*[30].

It has a simple geometric interpretation. The Fourier expansion of a function can be compared to the expansion of a vector $\mathbf{a}$ in an orthonormal basis $\mathbf{e}_n$ $(n = 1, \ldots, N)$, $\mathbf{a} = \sum_{n=1}^{N} a_n \mathbf{e}_n$, where a_n are the components of $\mathbf{a}$, and $\mathbf{e}_m \cdot \mathbf{e}_n = \delta_{nm}$. The square of $\mathbf{a}$ is then given by $\mathbf{a} \cdot \mathbf{a} = \sum_{n=1}^{N} a_n^2$, which is a finite-dimension vector analogue of Parseval's identity (3.29).

Example. The coefficients of the Fourier series for

$$f(x) = \begin{cases} \frac{\pi - x}{2}, & 0 < x < \pi, \\ -\frac{\pi + x}{2}, & -\pi < x < 0, \end{cases}$$

are $a_n = 0$, $b_n = 1/n$,

$$f(x) = \sum_{n=1}^{\infty} \frac{\sin nx}{n}.$$

Given that $\int_{-\pi}^{\pi} f^2(x)dx = 2\int_0^{\pi}\left(\frac{\pi - x}{2}\right)^2 dx = \frac{1}{2}\int_0^{\pi} x^2 dx = \pi^3/6$ (where we used substitution $\pi - x \to x$), Eq. (3.29) yields: $\pi^6/6 = \pi\sum_{n=1}^{\infty}(1/n)^2$, or

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Gibbs phenomenon. According to the convergence theorem, partial sums of the Fourier series for a piecewise smooth function approach the graph of $f(x)$ in every closed interval that does not contain the discontinuity, as the number of terms increases.

We illustrate this for

$$f(x) = \begin{cases} -\frac{\pi}{4}, & -\pi < x < 0, \\ \frac{\pi}{4}, & 0 < x < \pi. \end{cases} \tag{3.30}$$

which has the Fourier series

$$\sum_{m=0}^{\infty} \frac{\sin(2m+1)x}{2m+1}. \tag{3.31}$$

The figure below shows $f(x)$ together with the partial sums of its Fourier series containing 1, 2, 3, and 8 terms.

At the discontinuity the partial sums display oscillations which become progressively narrower and move closer to the discontinuity. However, they always "overshoot" (e.g., see the maximum nearest to the discontinuity shown by arrow on the graph), and the total oscillation in the approximating curve does not approach the jump in $f(x)$. This is known at the *Gibbs phenomenon*[31].

[30] Marc-Antoine Parseval des Chênes (1755–1836) was a French mathematician.

[31] Josiah Willard Gibbs (1839–1903) was an American scientist who made significant theoretical contributions to physics, chemistry, and mathematics. He discovered the "Gibbs

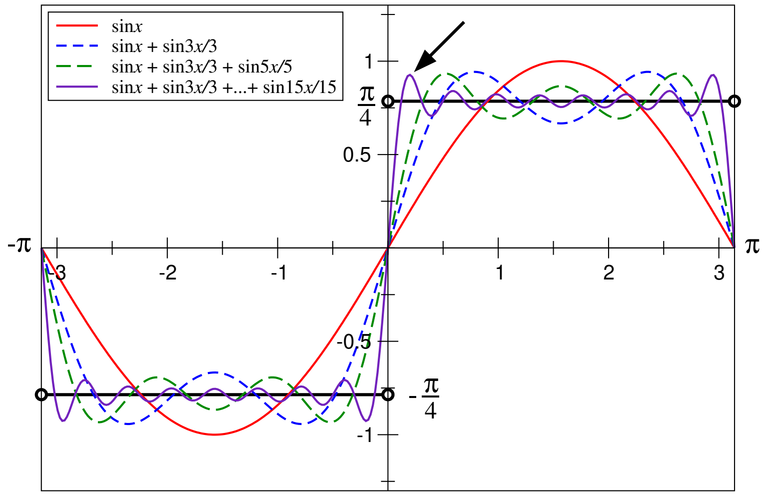


Figure 6: Partials Fourier series sums (3.32) for $f(x)$, for $N = 0$, 1, 2 and 7.

To analyse this effect, consider the partial sum of first $N+1$ terms of the Fourier series (3.31),

$$S_N(x) = \sum_{m=0}^{N} \frac{\sin(2m+1)x}{2m+1}. \tag{3.32}$$

Its derivative, $S'_N(x) = \sum_{m=0}^{N} \cos(2m+1)x$, is equal to the real part of the geometric series,

$$\sum_{m=0}^{N} e^{i(2m+1)x} = e^{ix}\,\frac{1-e^{i2(N+1)x}}{1-e^{i2x}} = \frac{1-e^{i2(N+1)x}}{e^{-ix}-e^{ix}} = \frac{e^{i2(N+1)x}-1}{2i\sin x},$$

so that

$$S'_N(x) = \mathrm{Re}\,\frac{e^{i2(N+1)x}-1}{2i\sin x} = \frac{\sin 2(N+1)x}{2\sin x}. \tag{3.33}$$

Let $x = x_m$ be the first maximum of $S_N(x)$ at $x > 0$. From $S'_N(x_m) = 0$, we find $x_m = \pi/[2(N+1)]$. Given that $S_N(0) = 0$, we can find the value of the partial sum at this maximum, $S_N(x_m)$, from the integral,

$$S_N(x_m) = \int_0^{x_m} S'_N(x)dx = \int_0^{x_m} \frac{\sin 2(N+1)x}{2\sin x}\,dx \simeq \int_0^{x_m} \frac{\sin 2(N+1)x}{2x}\,dx.$$

In the last expression we used $\sin x \simeq x$, which is valid for $x \ll 1$, and can be used for large N since $x_m \ll 1$. Using variable substitution $t = 2(N+1)x$ in the last integral, we obtain the partial sum at the maximum nearest to the discontinuity as

$$S_N(x_m) = \frac{1}{2}\int_0^{\pi} \frac{\sin t}{t}\,dt \approx 0.9260, \tag{3.34}$$

where the last value was obtained numerically. It is about 18% higher than $f(0+0) = \pi/4 \approx 0.7854$. This means that near the discontinuity the partial sums overshoot the graph of $f(x)$ by about 9% of the total size of the jump.

phenomenon" in 1899. However, as with many (if not most!) discoveries in mathematics, the "Gibbs phenomenon" was uncovered earlier by others, in this case, by Henry Wilbraham (1825–1883), an English mathematician who published a paper on the subject in 1848, see E. Hewitt and R. E. Hewitt, *The Gibbs-Wilbraham Phenomenon: An Episode in Fourier Analysis*, Archive for History of Exact Sciences, Vol. 21 (Springer, 1979).

4 Applications of the Fourier method

In this chapter we consider three examples of the use of the Fourier method, when solutions of the wave, heat and Laplace's equations are found using expansions in terms of the cosine or sine functions (i.e., Fourier series).

4.1 Vibrations of a string

Let us solve the problem of the motion of a string stretched between $x = 0$ and $x = l$, which is pulled to $u = h$ at $x = a$ and released from rest. The shape of the string at $t = 0$ is a piecewise linear function shown in Fig. 7.

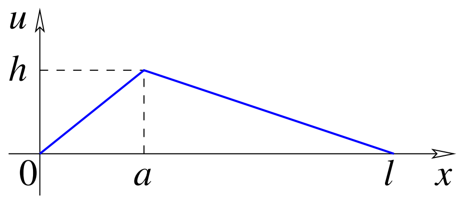


Figure 7: Initial shape of the string pulled to $u = h$ at $x = a$, Eq. (4.4).

The displacement of the string $u(x,t)$ satisfies the wave equation (Sec. 1.3),

$$u_{tt} - c^2 u_{xx} = 0, \tag{4.1}$$

with the boundary conditions

$$u(0,t) = 0, \quad u(l,t) = 0, \tag{4.2}$$

and the initial conditions

$$u(x,0) = f(x), \quad u_t(x,0) = 0, \tag{4.3}$$

where[32]

$$f(x) = \begin{cases} h\dfrac{x}{a}, & 0 \le x \le a, \\ h\dfrac{l-x}{l-a}, & a \le x \le l. \end{cases} \tag{4.4}$$

In Sec. 2.2, we used variable separation to solve Eq. (4.1) with boundary conditions (4.2). The solution we obtained, Eq. (2.12), can be written as

$$u(x,t) = (A\cos\omega_n t + B\sin\omega_n t)\sin\frac{n\pi x}{l}, \tag{4.5}$$

where $\omega_n = n\pi c/l$, $n = 1, 2, \ldots$, and A and B are arbitrary constants. Using the superposition principle (Sec. 1.6), we can construct a more general solution as a linear combination of solutions (4.5),

$$u(x,t) = \sum_{n=1}^{\infty}(A_n\cos\omega_n t + B_n\sin\omega_n t)\sin\frac{n\pi x}{l}, \tag{4.6}$$

[32] The expression for $f(x)$ is written easily, by inspection. For $0 \le x \le a$, $f(x)$ is a linear function through the origin which equals h for $x = a$.

where we relabelled the coefficients A_n and B_n, as they can be different for different n.

The coefficients A_n and B_n are determined by the initial conditions (4.3). Using the second of these first, we find the time derivative,

$$u_t(x,t) = \sum_{n=1}^{\infty}(-A_n\omega_n \sin\omega_n t + B_n\omega_n \cos\omega_n t)\sin\frac{n\pi x}{l},$$

and set it to zero for $t = 0$, which gives

$$\sum_{n=1}^{\infty} B_n\omega_n \sin\frac{n\pi x}{l} = 0.$$

The Fourier series on the left is zero for all x only if $B_n = 0$. Hence, Eq. (4.6) takes the form

$$u(x,t) = \sum_{n=1}^{\infty} A_n \cos\omega_n t \sin\frac{n\pi x}{l}. \tag{4.7}$$

Applying the first of the initial conditions (4.3), we have

$$\sum_{n=1}^{\infty} A_n \sin\frac{n\pi x}{l} = f(x).$$

The coefficients of this half-range Fourier series on $x \in [0, l]$ are determined using the formulae from Sec. 3.5 [see Eq. (3.27)],

$$A_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx.$$

For $f(x)$ from Eq. (4.4), we split the integration interval into two, 0 to a and a to l, and obtain, using integration by parts,

$$\begin{aligned}
A_n &= \frac{2}{l}\int_0^a h\frac{x}{a}\sin\frac{n\pi x}{l}\,dx + \frac{2}{l}\int_a^l h\frac{l-x}{l-a}\sin\frac{n\pi x}{l}\,dx \\
&= \frac{2h}{l}\left\{\left[\frac{x}{a}\,\frac{l}{n\pi}\left(-\cos\frac{n\pi x}{l}\right)\right]_0^a + \frac{l}{n\pi a}\int_0^a \cos\frac{n\pi x}{l}\,dx\right. \\
&\qquad\left. + \left[\frac{l-x}{l-a}\,\frac{l}{n\pi}\left(-\cos\frac{n\pi x}{l}\right)\right]_a^l - \frac{l}{n\pi(l-a)}\int_a^l \cos\frac{n\pi x}{l}\,dx\right\} \\
&= \frac{2h}{l}\left\{\cancel{-\frac{l}{n\pi}\cos\frac{n\pi a}{l}} + 0 + \frac{l^2}{n^2\pi^2 a}\left[\sin\frac{n\pi x}{l}\right]_0^a\right. \\
&\qquad\left. + 0 + \cancel{\frac{l}{n\pi}\cos\frac{n\pi a}{l}} - \frac{l^2}{n^2\pi^2(l-a)}\left[\sin\frac{n\pi x}{l}\right]_a^l\right\} \\
&= \frac{2h}{l}\left\{\frac{l^2}{n^2\pi^2 a}\left[\sin\frac{n\pi a}{l} - 0\right] - \frac{l^2}{n^2\pi^2(l-a)}\left[\sin n\pi - \sin\frac{n\pi a}{l}\right]\right\} \\
&= \frac{2h}{l}\,\frac{l^2}{n^2\pi^2}\left(\frac{1}{a} + \frac{1}{l-a}\right)\sin\frac{n\pi a}{l} \\
&= \frac{2hl^2}{n^2\pi^2 a(l-a)}\sin\frac{n\pi a}{l}.
\end{aligned} \tag{4.8}$$

Substituting this expression for A_n and $\omega_n = n\pi c/l$ into Eq. (4.7) gives the answer

$$u(x,t) = \frac{2hl^2}{\pi^2 a(l-a)} \sum_{n=1}^{\infty} \frac{\sin \frac{n\pi a}{l}}{n^2} \sin \frac{n\pi x}{l} \cos \frac{n\pi ct}{l}, \tag{4.9}$$

where we have taken the common fact outside the sum.

If the string is plucked in the middle, $a = l/2$, the Fourier coefficient A_n, Eq. (4.8), becomes

$$A_n = \frac{8h}{n^2\pi^2} \sin \frac{n\pi}{2},$$

where

$$\sin \frac{n\pi}{2} = \begin{cases} 0, & n = 2m, \\ (-1)^m, & n = 2m+1. \end{cases}$$

Hence, we see that the coefficients A_n equal zero for even n and are given by

$$A_n = \frac{8h}{\pi^2} \frac{(-1)^m}{(2m+1)^2} \tag{4.10}$$

for odd $n = 2m+1$, $m = 0, 1, 2, \ldots$. The vibration of the string is thus described by the sum Eq. (4.9) in which only odd terms are present,

$$u(x,t) = \frac{8h}{\pi^2} \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)^2} \sin \frac{(2m+1)\pi x}{l} \cos \frac{(2m+1)\pi ct}{l}. \tag{4.11}$$

This means that for a string plucked in the middle, only odd harmonics will contribute to the vibration. The first three odd harmonics are shown in Fig. 8.

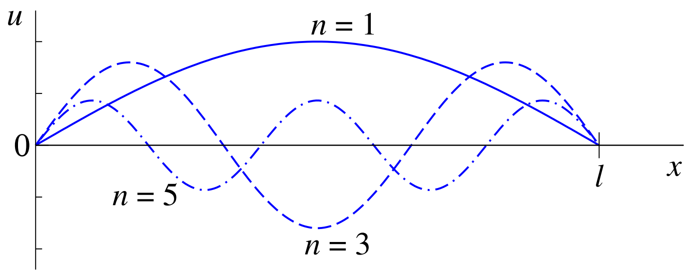


Figure 8: Shapes of the first three odd harmonics with arbitrary amplitudes.

4.2 Heat equation in one dimension

Let us now solve the heat equation (1.27),

$$u_t - K u_{xx} = 0, \tag{4.12}$$

for a rod of length l, whose ends are kept at zero temperature,

$$u(0,t) = 0, \quad u(l,t) = 0, \tag{4.13}$$

given the initial temperature distribution

$$u(x,0) = f(x). \tag{4.14}$$

We considered this problem in Sec. 2.4, and used variable separation to solve Eq. (4.12) with boundary conditions (4.13), to obtain solutions in the form (2.43),

$$u(x,t) = B_n \sin\frac{n\pi x}{l}\, e^{-(n^2\pi^2/l^2)Kt}, \tag{4.15}$$

where B_n is an arbitrary constant, and $n = 1, 2, \dots$.

Since Eq. (4.12) and the boundary conditions (4.13) are homogeneous, we can use the superposition principle to construct a more general solution in the form

$$u(x,t) = \sum_{n=1}^{\infty} B_n \sin\frac{n\pi x}{l}\, e^{-(n^2\pi^2/l^2)Kt}. \tag{4.16}$$

Setting $t = 0$ and applying the initial condition (4.14), we obtain

$$\sum_{n=1}^{\infty} B_n \sin\frac{n\pi x}{l} = f(x). \tag{4.17}$$

This a half-range Fourier sine series, cf. Eq. (3.26). Its coefficients are given by

$$B_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx. \tag{4.18}$$

Let us consider the initial temperature distribution shown in Fig. 9, i.e., a symmetric piecewise linear function with a maximum $u = T$ at $x = l/2$.

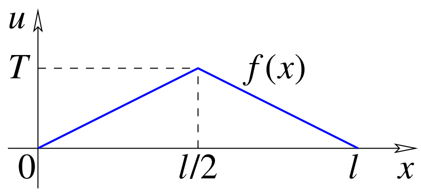


Figure 9: Initial temperature distribution for a rod of length l.

This function is the same as the initial shape of the string considered in Sec. 4.1, Eq. (4.4), in which we replace h by T and use $a = l/2$. Hence, we can use the expression obtained earlier, Eq. (4.10), and have

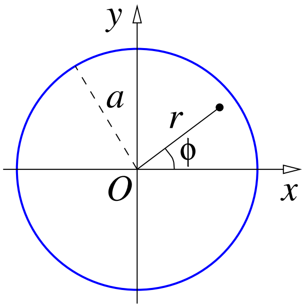
$$B_n = \frac{8T}{\pi^2}\frac{(-1)^m}{(2m+1)^2} \tag{4.19}$$

for odd $n = 2m + 1$, $m = 0, 1, 2, \dots$, with $B_n = 0$ for even n. Substituting this into Eq. (4.16) and changing the summation index from n to m gives the final answer

$$u(x,t) = \frac{8T}{\pi^2}\sum_{m=0}^{\infty}\frac{(-1)^m}{(2m+1)^2}\sin\frac{(2m+1)\pi x}{l}\, e^{-(2m+1)^2\pi^2 Kt/l^2}. \tag{4.20}$$

Note that for $t \to \infty$, $e^{-(2m+1)^2\pi^2 Kt/l^2} \to 0$. Therefore, $u(x,t) \to 0$ as $t \to \infty$, so the rod cools down to the temperature of its ends.

4.3 Laplace's equation for a disk



In this section we use the Fourier series to solve Laplace's equation, Eq. (1.28),

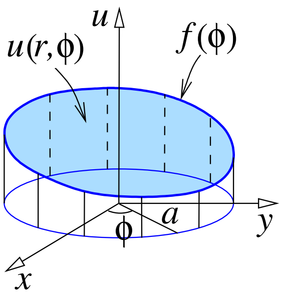
$$\nabla^2 u = 0, \tag{4.21}$$

in two dimensions, for a disk of radius a, assuming that the values of u are fixed on the boundary (i.e., on the circle of radius a).

In this problem, it is convenient to use the plane polar coordinates (r, ϕ). The boundary condition for $u(r, \phi)$ can then be written as

$$u(a, \phi) = f(\phi), \tag{4.22}$$

where $f(\phi)$ is a given function.



To see what this problem describes, recall the wave equation (Sec. 1.3)

$$u_{tt} - c^2 \nabla^2 u = 0,$$

and the heat equation (Sec. 1.4)

$$u_t - K \nabla^2 u = 0.$$

For time-independent, *stationary* solutions of these equations, the time derivatives u_t and u_{tt} vanish, hence, these solutions satisfy $\nabla^2 u = 0$. Therefore, the problem for the disk posed above, describes the shape of a drum skin stretched on a circular drum with an uneven edge whose hight is given by the function $f(\phi)$. It also gives the two-dimensional temperature distribution in a circular room, in which the temperature of the wall is described by $f(\phi)$.

Laplace's equation in plane polar coordinates reads (cf. Sec. 2.3),

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \phi^2} = 0. \tag{4.23}$$

Using variable separation, one finds its solution in the form[33]

$$u(r,\phi) = C \ln r + D + \sum_{n=1}^{\infty} r^n (A_n \cos n\phi + B_n \sin n\phi) + \sum_{n=1}^{\infty} r^{-n} (\tilde{A}_n \cos n\phi + \tilde{B}_n \sin n\phi),$$

where C, D, A_n, B_n, $\tilde{A}_n$, and $\tilde{B}_n$ are arbitrary coefficients. The solution $u(r, \phi)$ must be nonsingular (i.e., finite) at the origin, $r = 0$, hence we set $C = 0$ and $\tilde{A}_n = \tilde{B}_n = 0$, and write the solution in the form

$$u(r,\phi) = \frac{A_0}{2} + \sum_{n=1}^{\infty} r^n (A_n \cos n\phi + B_n \sin n\phi), \tag{4.24}$$

where we also set $D = A_0/2$, to fit the general form of the Fourier series (3.2).

Using Eq. (4.24) in the boundary condition, we have

$$\frac{A_0}{2} + \sum_{n=1}^{\infty} a^n (A_n \cos n\phi + B_n \sin n\phi) = f(\phi).$$

[33] See Example 1 in Problem sheet 7.

This a Fourier series with the coefficients A_0, $a^n A_n$ and $a^n B_n$, which we find using the standard formulae (3.3):

$$A_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\phi)d\phi,$$
$$a^n A_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\phi)\cos n\phi\, d\phi,$$
$$a^n B_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\phi)\sin n\phi\, d\phi.$$

The last two equations give

$$A_n = \frac{1}{\pi}\frac{1}{a^n}\int_{-\pi}^{\pi} f(\phi)\cos n\phi\, d\phi, \quad B_n = \frac{1}{\pi}\frac{1}{a^n}\int_{-\pi}^{\pi} f(\phi)\sin n\phi\, d\phi.$$

The solution $u(r,\phi)$ is now found by substituting the above expressions for A_0, A_n and B_n into Eq. (4.24). However, since ϕ appears as an independent variable in Eq. (4.24), we change the integration variable from ϕ to ψ in the integrals, and obtain

$$u(r,\phi) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(\psi)d\psi + \frac{1}{\pi}\sum_{n=1}^{\infty}\frac{r^n}{a^n}\left(\int_{-\pi}^{\pi} f(\psi)\cos n\psi\, d\psi\, \cos n\phi \right.$$
$$\left. + \int_{-\pi}^{\pi} f(\psi)\cos n\psi\, d\psi\, \sin n\phi\right),$$

which can be written as

$$u(r,\phi) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(\psi)d\psi + \frac{1}{\pi}\sum_{n=1}^{\infty}\frac{r^n}{a^n}\int_{-\pi}^{\pi} f(\psi)(\cos n\psi\cos n\phi + \sin n\psi\sin n\phi)d\psi.$$

Using $\cos n\psi\cos n\phi + \sin n\psi\sin n\phi = \cos(\psi-\phi)$, interchanging the order of summation and integration, and taking out some common factors, we have

$$u(r,\phi) = \frac{1}{2\pi}\int_{-\pi}^{\pi}\left[1 + 2\sum_{n=1}^{\infty}\frac{r^n}{a^n}\cos n(\psi-\phi)\right] f(\psi)d\psi. \tag{4.25}$$

The sum over n is similar to that of a geometric series, except that it contains an "awkward" $\cos n(\psi-\phi)$ factor, and that it starts from $n=1$, instead of the more common $n=0$. To overcome the first problem, we use the relation $\cos\theta = \frac{1}{2}(e^{i\theta}+e^{-i\theta})$ and also add and subtract the $n=0$ terms. The expression in square brackets then becomes

$$1 + 2\sum_{n=1}^{\infty}\frac{r^n}{a^n}\cos n(\psi-\phi) = 1 + \sum_{n=1}^{\infty}\frac{r^n}{a^n}\left(e^{in(\psi-\phi)} + e^{-in(\psi-\phi)}\right)$$
$$= 1 + \sum_{n=0}^{\infty}\left(\frac{r}{a}e^{i(\psi-\phi)}\right)^n + \sum_{n=0}^{\infty}\left(\frac{r}{a}e^{-i(\psi-\phi)}\right)^n - 2$$

Using the knows expression for the sum of the infinite geometric series, $S = 1 + x + x^2 + \cdots = 1/(1-x)$[34], we obtain

[34] To verify this, we consider the infinite series $S = 1 + x + x^2 + x^3 + \ldots$, and take out x as the common factor in all terms past the first one, $S = 1 + x(1 + x + x^2 + \ldots)$, which gives $S = 1 + xS$ and leads to $S = 1/(1-x)$.

$$
\begin{aligned}
1+2\sum_{n=1}^{\infty}\frac{r^n}{a^n}\cos n(\psi-\phi) &= \frac{1}{1-\dfrac{r}{a}e^{i(\psi-\phi)}}+\frac{1}{1-\dfrac{r}{a}e^{-i(\psi-\phi)}}-1 \\
&= \frac{1-\dfrac{r}{a}e^{-i(\psi-\phi)}+1-\dfrac{r}{a}e^{i(\psi-\phi)}}{\left(1-\dfrac{r}{a}e^{i(\psi-\phi)}\right)\left(1-\dfrac{r}{a}e^{-i(\psi-\phi)}\right)}-1 \\
&= \frac{2-2\dfrac{r}{a}\cos(\psi-\phi)}{1-2\dfrac{r}{a}\cos(\psi-\phi)+\dfrac{r^2}{a^2}}-1 \\
&= \frac{1-\dfrac{r^2}{a^2}}{1-2\dfrac{r}{a}\cos(\psi-\phi)+\dfrac{r^2}{a^2}} \\
&= \frac{a^2-r^2}{a^2-2ar\cos(\psi-\phi)+r^2}.
\end{aligned}
$$

Substituting the last expression into Eq. (4.25) gives the answer in the form

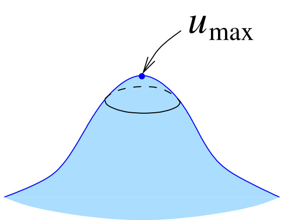
$$
u(r,\phi)=\frac{1}{2\pi}\int_{-\pi}^{\pi}\frac{(a^2-r^2)f(\psi)d\psi}{a^2-2ar\cos(\psi-\phi)+r^2}, \tag{4.26}
$$

known as *Poisson's integral*.[35]

In particular, Eq. (4.26) shows that the value of u at the origin $r=0$, is

$$
u(0,\phi)=\frac{1}{2\pi}\int_{-\pi}^{\pi}f(\psi)d\psi. \tag{4.27}
$$

The integral on the right-hand side is the average of u over the circle $r=a$ [cf. Eq. (4.22)]. This means that solutions of Laplace's equation in a given domain cannot have minima or maxima inside the domain.



To prove this, let us assume to the contrary that u has a maximum with $u=u_{\max}$. This means that the values of u in a certain vicinity of the maximum point (but away from it) are lower, i.e., $u<u_{\max}$. However, by Eq. (4.27), the value of u at any point must be equal to the average of its values over a circle centred on this point. If this point is at the maximum, the integral will give a value smaller than $u_{\max}$, hence, we have a contradiction.

It is instructive to check that the solution in the form of Poisson's integral, Eq. (4.26), satisfies the boundary condition (4.22), i.e., that $u(a,\phi)=f(\phi)$. At first sight, setting $r=a$ under the integral makes it equal to zero. However, one may notice that for $\psi=\phi$, the denominator equals $a^2-2ar+r^2=(a-r)^2$, so it is also zero for $r=a$. Hence, we need to consider the limit $r\to a-0$ for small $|\psi-\phi|$ more carefully.

To simplify the task, we introduce ξ such that $a-r=\xi a$, or $r=a(1-\xi)$. The limit $r\to a$ is then equivalent to $\xi\to 0$. We also use the Taylor expansion of the cosine function[36] to approximate $\cos(\psi-\phi)\simeq 1-\frac{1}{2}(\psi-\phi)^2$. Using this,

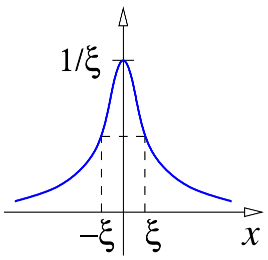
[35] Note that the denominator is the squared distance between points (r,ϕ) and (a,ψ).

[36] Recall the Taylor (or Maclaurin) expansion, $\cos x = 1-\dfrac{x^2}{2}+\dfrac{x^4}{4!}-\dfrac{x^6}{6!}+\dots$.

we obtain for the factor that multiplies $f(\psi)$ in Eq. (4.26),

$$\begin{aligned}\frac{1}{2\pi}\frac{(a^2-r^2)}{a^2-2ar\cos(\psi-\phi)+r^2} &= \frac{1}{2\pi}\frac{a^2-a^2(1-\xi)^2}{a^2-2a^2(1-\xi)\cos(\psi-\phi)+a^2(1-\xi)^2}\\ &= \frac{1}{2\pi}\frac{1-(1-\xi)^2}{1-2(1-\xi)\cos(\psi-\phi)+(1-\xi)^2}\\ &\simeq \frac{1}{2\pi}\frac{2\xi-\xi^2}{2-2(1-\xi)\left[1-\frac{1}{2}(\psi-\phi)^2\right]-2\xi+\xi^2}\\ &\simeq \frac{1}{2\pi}\frac{2\xi}{(\psi-\phi)^2+\xi^2}\\ &= \frac{1}{\pi}\frac{\xi}{(\psi-\phi)^2+\xi^2},\end{aligned}$$

where we neglected ξ^2 compared with 2ξ in the numerator (since ξ is small), and also neglected the product $\xi(\psi-\phi)^2$ in the denominator, as both ξ and $\psi-\phi$ are regarded as small, making this product "extra small".



Looking at the expression

$$\frac{\xi}{x^2+\xi^2},$$

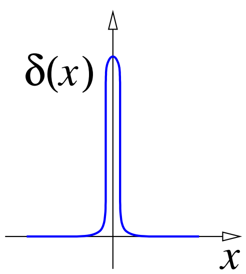
as a function of x, we see that it is even and has a maximum at $x=0$, where it equals $1/\xi$. We also see that this function decreases to 1/2 of its maximum value at $x=\pm\xi$. As ξ tends to zero, this function becomes taller but narrower, while the area under the graph remains the same:

$$\int_{-\infty}^{\infty}\frac{\xi}{x^2+\xi^2}dx = \int_{-\infty}^{\infty}\frac{1}{1+(x/\xi)^2}d(x/\xi) = \left[\arctan\frac{x}{\xi}\right]_{-\infty}^{\infty} = \frac{\pi}{2}-\left(-\frac{\pi}{2}\right) = \pi.$$

For the function

$$\frac{1}{\pi}\frac{\xi}{x^2+\xi^2},$$

the area under its graph equals unity. In the limit $\xi\to 0$, this function becomes infinitely tall but infinitely "slim", and turns into the *Dirac delta function* $\delta(x)$, defined as follows[37]



$$\delta(x) = \begin{cases} 0, & x\neq 0,\\ \infty, & x=0,\end{cases} \tag{4.28}$$

$$\int_{-\infty}^{\infty}\delta(x)dx = 1. \tag{4.29}$$

Note that the integral in Eq. (4.29) equals unity for *any* interval that contains $x=0$, i.e., $\int_\alpha^\beta \delta(x)dx = 1$ for $\alpha<0<\beta$, since $\delta(x)=0$ outside this interval.

The main property of the delta function is

$$\int_{-\infty}^{\infty} f(x)\delta(x-a)dx = f(a), \tag{4.30}$$

[37] The delta function was introduced in 1930 by Paul Dirac (1902-1984) as a useful tool in Quantum Mechanics. Dirac's definition was not rigorous by pure mathematics standards. Subsequently, it was introduced rigorously in the theory of generalised functions or distributions, pioneered by Sergei Sobolev (1908–1989) and Laurent Schwartz (1915–2002).

where $f(x)$ is continuous at $x = a$. To verify this, note that the only value of x that contributes to the integral is $x = a$ where $f(x) = f(a)$. Hence, we have

$$\int_{-\infty}^{\infty} f(x)\delta(x-a)dx = \int_{-\infty}^{\infty} f(a)\delta(x-a)dx = f(a)\int_{-\infty}^{\infty} \delta(x-a)dx = f(a).$$

Note also that Eq. (4.30) will hold for the integral over any interval that contains a, e.g., $\int_{a-\varepsilon}^{a+\varepsilon} f(x)\delta(x-a)dx = f(a)$ for any $\varepsilon > 0$.

Going back to Eq. (4.26), we now see that for $r \to a - 0$,

$$\frac{1}{2\pi}\frac{(a^2 - r^2)}{a^2 - 2ar\cos(\psi - \phi) + r^2} \longrightarrow \delta(\psi - \phi), \tag{4.31}$$

Therefore, in the limit $r \to a - 0$, the right-hand side of Eq. (4.26) gives

$$\int_{-\pi}^{\pi} \delta(\psi - \phi) f(\psi) d\psi = f(\phi), \tag{4.32}$$

i.e., as expected, the function $u(r, \phi)$ satisfies the boundary condition[38].

We will see the Dirac delta function again in Sec. 5.1 and Sec. 5.3.

[38] The expression on the left-hand side of (4.31) is a periodic function of $\psi - \phi$ with the period 2π. Hence, its limit for $r \to a - 0$ is, strictly speaking, equal to $\sum_{n=-\infty}^{n=+\infty} \delta(\psi - \phi + 2\pi n)$. However, on $[\pi, \pi]$, only the $n = 0$ term in this sum contributes.

5 Fourier transform

5.1 Fourier integral

The Fourier integral allows one to represent $f(x)$ on the whole real axis.

Consider the Fourier series for $f(x)$ on $-l \leq x \leq l$ in complex form,[39]

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/l}, \tag{5.1}$$

$$c_n = \frac{1}{2l} \int_{-l}^{l} e^{-in\pi x/l} f(x) dx, \tag{5.2}$$

Taken together, these equation give

$$f(x) = \frac{1}{2l} \sum_{n=-\infty}^{\infty} e^{in\pi x/l} \int_{-l}^{l} e^{-in\pi\xi/l} f(\xi) d\xi. \tag{5.3}$$

Introducing $p_n = n\pi/l$ and $\Delta p = p_{n+1} - p_n = \pi/l$, we re-write (5.3) as

$$f(x) = \frac{1}{2\pi} \sum_{p_n=-\infty}^{\infty} e^{ip_n x} \Delta p \int_{-l}^{l} e^{-ip_n \xi} f(\xi) d\xi. \tag{5.4}$$

The sum in (5.4) is a Riemann sum. In the limit $l \to \infty$, $\Delta p \to 0$, it becomes an integral over p, and we have

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ipx} dp \int_{-\infty}^{\infty} e^{-ip\xi} f(\xi) d\xi. \tag{5.5}$$

Commonly, this relation is written with p replaced by $-p$, as a combination of two formulae,

$$F(p) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ipx} f(x) dx, \tag{5.6}$$

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ipx} F(p) dp. \tag{5.7}$$

Here $F(p) \equiv \mathcal{F}[f]$ is the *Fourier transform* of $f(x)$, and $f(x) \equiv \mathcal{F}^{-1}[F]$ is the inverse Fourier transform of $F(p)$.

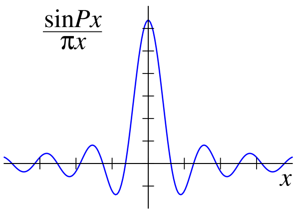
To verify these relations, we substitute (5.6) (with x changed to ξ, to avoid confusion) into (5.7), and make the integration limits in (5.7) finite, e.g., $-P$ and P,

$$\frac{1}{2\pi} \int_{-P}^{P} e^{-ipx} dp \int_{-\infty}^{\infty} e^{ip\xi} f(\xi) d\xi, \tag{5.8}$$

[39] Compare (3.2), (3.3) and (3.22), (3.23) with (3.15), (3.16).

and will later take the limit $P \to \infty$. Changing the order of integration in (5.8), we obtain

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} f(\xi)d\xi \int_{-P}^{P} e^{ip(\xi-x)}dp = \int_{-\infty}^{\infty} f(\xi)\frac{\sin P(\xi-x)}{\pi(\xi-x)}d\xi. \tag{5.9}$$



The function

$$\frac{\sin P(\xi-x)}{\pi(\xi-x)} \tag{5.10}$$

in the integrand is even, and has a maximum at $\xi \to x$, where it equals P/π. The "width" of this maximum is $|\xi - x| \sim \pi/P$. One can also show that

$$\int_0^{\infty} \frac{\sin Ps}{s}ds = \int_0^{\infty} \frac{\sin s}{s}ds = \frac{\pi}{2}, \tag{5.11}$$

which is known as the *Dirichlet integral*.[40] This means that in the limit $P \to \infty$, (5.10) becomes the δ-function,

$$\lim_{P\to\infty} \frac{\sin P(\xi-x)}{\pi(\xi-x)} = \delta(\xi - x). \tag{5.12}$$

Hence, taking the limit $P \to \infty$ of the right-hand side of (5.9), we obtain

$$\int_{-\infty}^{\infty} f(\xi)\delta(\xi-x)d\xi = f(x),$$

as required. Formally, this result proves the following identity:

$$\int_{-\infty}^{\infty} e^{ip(\xi-x)}dp = 2\pi\delta(\xi - x). \tag{5.13}$$

Example 1. Let us find the Fourier transform of $f(x) = e^{-\alpha|x|}$.

$$\begin{aligned} F(p) &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{ipx}e^{-\alpha|x|}dx \\ &= \frac{1}{\sqrt{2\pi}}\left[\int_{-\infty}^{0} e^{(\alpha+ip)x}dx + \int_0^{\infty} e^{-(\alpha-ip)x}dx\right] \\ &= \frac{1}{\sqrt{2\pi}}\left[\frac{1}{\alpha+ip} + \frac{1}{\alpha-ip}\right] = \frac{1}{\sqrt{2\pi}}\frac{2\alpha}{\alpha^2+p^2}, \end{aligned}$$

where we used $|x| = -x$ for $x < 0$ and $|x| = x$ for $x \geq 0$, to get rid of the modulus sign. Hence,

$$\mathcal{F}\left[e^{-\alpha|x|}\right] = \frac{1}{\sqrt{2\pi}}\frac{2\alpha}{\alpha^2+p^2}.$$

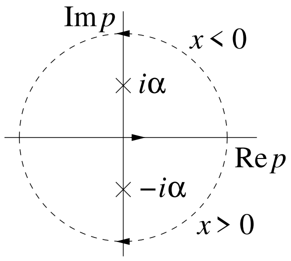
[40] The integral $\int_0^\infty \frac{\sin Ps}{s}ds$ is an odd function of P, and it equals zero for $P = 0$. For $P > 0$, $\int_0^\infty \frac{\sin Ps}{s}ds = \int_0^\infty \frac{\sin Ps}{Ps}d(Ps) = \int_0^\infty \frac{\sin s}{s}ds$. Using $\frac{1}{s} = \int_0^\infty e^{-su}du$ under the integral, writing $\sin s = \operatorname{Im} e^{is}$ and changing the order of integration, we have:

$$\int_0^\infty \frac{\sin s}{s}ds = \int_0^\infty \operatorname{Im}\left[\int_0^\infty e^{-(u-i)s}ds\right]du = \int_0^\infty \operatorname{Im}\left[\frac{1}{u-i}\right]du = \int_0^\infty \frac{du}{1+u^2} = \frac{\pi}{2}.$$

The inverse Fourier transform

$$\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-ipx}\frac{2\alpha}{\sqrt{2\pi}(\alpha^2+p^2)}\,dp = \frac{i}{2\pi}\int_{-\infty}^{\infty} e^{-ipx}\left[\frac{1}{p+i\alpha}-\frac{1}{p-i\alpha}\right]dp,$$

is found by integrating in the complex p plane using Cauchy's formula [41].



The integral above is from $-\infty$ to $+\infty$ along the real axis of p. For $x>0$ we can close this integration path with a semicircle in the lower half-plane of p (since $e^{-ipx}=e^{-i\mathrm{Re}\,p\,x}e^{\mathrm{Im}\,p\,x}$ decreases into the lower half-plane ($\mathrm{Im}\,p<0$), and when the radius of the semicircle tends to infinity, the integral over it vanishes and does not change the value of the integral). This contour encloses the pole $p=-i\alpha$ of the first term in square brackets, which gives (by Cauchy's formula)

$$\frac{i}{2\pi}(-1)2\pi i e^{-i(-i\alpha)x} = e^{-\alpha x}, \tag{5.14}$$

where the extra minus sign is due to the contour being traversed in the clockwise direction. For $x<0$ the contour can be closed in the upper half-plane. This contour will contain a pole at $p=i\alpha$ from the second term in brackets. Cauchy's formula then yields

$$\frac{i}{2\pi}\left[-2\pi i e^{-i(i\alpha)x}\right] = e^{\alpha x}. \tag{5.15}$$

The two answers (5.14) and (5.15) can be written as a single formula using $|x|$, so we have (as expected)

$$\mathcal{F}^{-1}\left[\frac{1}{\sqrt{2\pi}}\frac{2\alpha}{\alpha^2+p^2}\right] = e^{-\alpha|x|}.$$

Example 2. Find the Fourier transform of $f(x)=e^{-x^2/2}$. To find

$$F(p) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{ipx}e^{-x^2/2}dx, \tag{5.16}$$

we use the following standard integral,[42]

$$\int_{-\infty}^{\infty} e^{-\alpha x^2+\beta x}dx = \sqrt{\frac{\pi}{\alpha}}e^{\beta^2/4\alpha}. \tag{5.17}$$

[41] If $f(z)$ is analytic everywhere within and on a simple closed curve C and z_0 is inside C, then $\oint \frac{f(z)}{z-z_0}dz = 2\pi i f(z_0)$, where the integral is taken in the positive direction.

[42] First, consider $I=\int_{-\infty}^{\infty} e^{-x^2}dx$. Its square is found using polar coordinates ρ and ϕ:

$$I^2 = \int_{-\infty}^{\infty} e^{-x^2}dx\int_{-\infty}^{\infty} e^{-y^2}dy = \iint_{-\infty}^{\infty} e^{-(x^2+y^2)}dxdy = \int_0^{2\pi}\int_0^{\infty} e^{-\rho^2}\rho d\rho d\phi = \frac{2\pi}{2}\int_0^{\infty} e^{-\rho^2}d(\rho^2) = \pi.$$

Hence, $I=\sqrt{\pi}$ and $\int_{-\infty}^{\infty} e^{-\alpha x^2}dx = \sqrt{\pi/\alpha}$, obtained by changing variable to $s=\sqrt{\alpha}x$. Finally, Eq. (5.17) is obtained by completing the square in the exponent,

$$\int_{-\infty}^{\infty} e^{-\alpha x^2+\beta x}dx = \int_{-\infty}^{\infty} e^{-\alpha[(x-\beta/2\alpha)^2-\beta^2/4\alpha^2]}dx = e^{\beta^2/4\alpha}\int_{-\infty}^{\infty} e^{-\alpha(x-\beta/2\alpha)^2}dx = e^{\beta^2/4\alpha}\sqrt{\frac{\pi}{\alpha}}.$$

For the integral (5.16) ($\alpha = \frac{1}{2}$, $\beta = ip$) this gives

$$F(p) = \frac{1}{\sqrt{2\pi}}\sqrt{2\pi}e^{(ip)^2/2} = e^{-p^2/2}.$$

Hence, $\mathcal{F}\left[e^{-x^2/2}\right] = e^{-p^2/2}$, and similarly $\mathcal{F}^{-1}\left[e^{-p^2/2}\right] = e^{-x^2/2}$.

5.2 Fourier cosine and sine transforms

Let us assume that $f(x)$ is an even function. Using Euler's formula, one obtains

$$\begin{aligned} F(p) &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{ipx} f(x)dx \\ &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} (\cos px + i\sin px) f(x)dx \\ &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \cos px\, f(x)dx + \frac{i}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \sin px\, f(x)dx. \end{aligned} \tag{5.18}$$

The integrand in the first integral above is even, and this integral is twice that from 0 to ∞. The integrand containing $\sin px$ is an odd function, and the second integral in (5.18) is zero.

Hence we define the *Fourier cosine transform*,

$$F_c(p) \equiv \mathcal{F}_c[f] = \sqrt{\frac{2}{\pi}}\int_0^{\infty} f(x)\cos px\, dx. \tag{5.19}$$

Since $\cos px$ is even, $F_c(p)$ is even, and the inverse Fourier cosine transform is

$$f(x) = \mathcal{F}_c^{-1}[F_c] = \sqrt{\frac{2}{\pi}}\int_0^{\infty} F_c(p)\cos px\, dp. \tag{5.20}$$

For an odd function one can similarly introduce the *Fourier sine transform*[43]

$$F_s(p) \equiv \mathcal{F}_s[f] = \sqrt{\frac{2}{\pi}}\int_0^{\infty} f(x)\sin px\, dx, \tag{5.21}$$

with the inverse

$$f(x) = \mathcal{F}_s^{-1}[F_s] = \sqrt{\frac{2}{\pi}}\int_0^{\infty} F_s(p)\sin px\, dp. \tag{5.22}$$

Both the Fourier cosine and Fourier sine transforms can be applied to functions defined on $0 \le x < \infty$.

[43] As seen from Eq. (5.18), in this case the first integral vanishes. The second integral contains a factor i, which makes the transform of a real function imaginary. By convention, this factor i is dropped in both the direct and inverse Fourier sine transform formulae.

5.3 Application to the heat equation

Fourier transforms can be used to solve PDE on infinite or semi-infinite intervals.

Problem 1. Solve the *heat equation*, $u_t - Ku_{xx} = 0$ on $-\infty < x < \infty$, with the initial condition $u(x,0) = f(x)$ and boundary conditions $u, u_x \to 0$ for $x \to \pm\infty$.

Let us introduce the Fourier transform of $u(x,t)$ with respect to x,

$$\mathcal{F}[u] \equiv U(p,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ipx} u(x,t) dx. \tag{5.23}$$

This gives the following expressions for the Fourier transforms of the *derivatives* of $u(x,t)$:

$$\mathcal{F}[u_t] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ipx} \frac{\partial u}{\partial t} dx = \frac{\partial}{\partial t} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ipx} u(x,t) dx = \frac{\partial U(p,t)}{\partial t}, \tag{5.24}$$

as integration over x and differentiation with respect to t are interchangeable. For the x derivative we integrate by parts twice,

$$\begin{aligned} \mathcal{F}[u_{xx}] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ipx} \frac{\partial^2 u}{\partial x^2} dx \\ &= \frac{1}{\sqrt{2\pi}} \left(\underbrace{\left[e^{ipx} \frac{\partial u}{\partial x} \right]_{-\infty}^{+\infty}}_{=0} - ip \int_{-\infty}^{\infty} e^{ipx} \frac{\partial u}{\partial x} dx \right) \\ &= \frac{1}{\sqrt{2\pi}} \left(\underbrace{\left[-ipe^{ipx} u(x,t) \right]_{-\infty}^{+\infty}}_{=0} + (ip)^2 \int_{-\infty}^{\infty} e^{ipx} u(x,t) dx \right) \\ &= -p^2 U(p,t), \end{aligned} \tag{5.25}$$

where we used the boundary conditions at $x \to \pm\infty$.

Hence, taking the Fourier transform of the heat equation, we have

$$\frac{\partial U}{\partial t} + Kp^2 U(p,t) = 0.$$

This is in fact an *ordinary* differential equation for $U(p,t)$ as a function of t, with p being a parameter. Its general solution is

$$U(p,t) = C(p) e^{-Kp^2 t}, \tag{5.26}$$

where $C(p)$ is an arbitrary "constant" (i.e., an arbitrary function of parameter p). For $t = 0$, Eq. (5.26) gives $U(p,0) = C(p)$. On the other hand, from the initial condition, we have

$$U(p,0) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ipx} u(x,0) dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ipx} f(x) dx = F(p), \tag{5.27}$$

where $F(p) = \mathcal{F}[f]$. Hence, $C(p) = F(p)$ and we have $U(p,t) = F(p) e^{-Kp^2 t}$.

The solution $u(x,t)$ is found by performing the inverse Fourier transform,

$$\begin{aligned} u(x,t) &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-ipx}U(p,t)dp \\ &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-ipx}F(p)e^{-Kp^2t}dp \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-ipx}\int_{-\infty}^{\infty} e^{-ip\xi}f(\xi)d\xi e^{-Kp^2t}dp, \end{aligned}$$

where we have changed the integration variable x to ξ when substituting the expression for $F(p)$, Eq. (5.27), to avoid confusion.

Chaging the order of integration and using Eq. (5.17) to integrate over p [with $\alpha = Kt$ and $\beta = i(\xi - x)$], we obtain

$$\begin{aligned} u(x,t) &= \frac{1}{2\pi}\int_{-\infty}^{\infty} f(\xi)\int_{-\infty}^{\infty} e^{-Kp^2t+i(\xi-x)p}dpd\xi \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty} f(\xi)\sqrt{\frac{\pi}{Kt}}e^{-(\xi-x)^2/4Kt}d\xi, \end{aligned}$$

which gives the final answer:

$$u(x,t) = \int_{-\infty}^{\infty} \frac{e^{-(\xi-x)^2/4Kt}}{\sqrt{4\pi Kt}} f(\xi)d\xi. \tag{5.28}$$

Note that for $t \to 0$ the left-hand side becomes $f(x)$, which means that

$$\lim_{t\to 0} \frac{e^{-(\xi-x)^2/4Kt}}{\sqrt{4\pi Kt}} = \delta(\xi - x),$$

This can be proved directly by examining the behaviour of this function for $t \to 0$, and proving that its integral equals unity.

5.4 Application to the wave equation

Let us solve the one-dimensional *wave equation* for $u(x,t)$ on $-\infty < x < \infty$,

$$u_{tt} - c^2 u_{xx} = 0, \tag{5.29}$$

with the initial conditions $u(x,0) = f(x)$, $u_t(x,0) = g(x)$.

Let $U(p,t) = \mathcal{F}[u]$ be the Fourier transform of $u(x,t)$ with respect to x. Then, similar to Eqs. (5.24) and (5.25),

$$\mathcal{F}[u_{tt}] = \frac{\partial^2 U}{\partial t^2}, \tag{5.30}$$

$$\mathcal{F}[u_{xx}] = -p^2 U(p,t), \tag{5.31}$$

where we assume $u(x,t)$, $u_x(x,t) \to 0$ for $x \to \pm\infty$ in (5.31). Fourier-transforming Eq. (5.29) gives

$$\frac{\partial^2 U}{\partial t^2} + p^2c^2U(p,t) = 0. \tag{5.32}$$

This is an *ordinary* differential equation, p being a parameter. Its general solution is

$$U(p,t) = A(p)\cos pct + B(p)\sin pct, \tag{5.33}$$

where $A(p)$ and $B(t)$ are arbitrary functions. To find them, we Fourier-transform the initial conditions, which gives

$$U(p,0) = F(p), \tag{5.34}$$

$$\left.\frac{\partial U}{\partial t}\right|_{t=0} = G(p), \tag{5.35}$$

where $F(p) = \mathcal{F}[f]$ and $G(p) = \mathcal{F}[g]$. On the other hand, from Eq. (5.33),

$$U(p,0) = A(p), \tag{5.36}$$

$$\left.\frac{\partial U}{\partial t}\right|_{t=0} = pcB(p), \tag{5.37}$$

Comparing these equations with (5.34) and (5.35), we find $A(p) = F(p)$ and $B(p) = G(p)/pc$, so that

$$U(p,t) = F(p)\cos pct + \frac{G(p)}{pc}\sin pct. \tag{5.38}$$

To find $u(x,t)$ we using the inverse Fourier transform,

$$\begin{aligned}
u(x,t) &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty} e^{-ipx}\left[F(p)\cos pct + \frac{G(p)}{pc}\sin pct\right]dp \\
&= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\frac{e^{ipct}+e^{-ipct}}{2}e^{-ipx}F(p)dp \\
&\quad + \frac{1}{c\sqrt{2\pi}}\int_{-\infty}^{\infty}\frac{e^{ipct}-e^{-ipct}}{2i}e^{-ipx}\frac{G(p)}{p}dp \\
&= \frac{1}{2\sqrt{2\pi}}\int_{-\infty}^{\infty}\left[e^{-ip(x-ct)}+e^{-ip(x+ct)}\right]F(p)dp \\
&\quad + \frac{1}{2c\sqrt{2\pi}}\int_{-\infty}^{\infty}\frac{e^{-ip(x-ct)}-e^{-ip(x+ct)}}{ip}G(p)dp.
\end{aligned} \tag{5.39}$$

The integral contaning $F(p)$ is the sum of two originals $f(x)$, shifted by ct and $-ct$, while the term containing $G(p)$ is a definite integral of $g(x)$,[44] so that

$$u(x,t) = \frac{1}{2}[f(x-ct)+f(x+ct)] + \frac{1}{2c}\int_{x-ct}^{x+ct} g(\xi)d\xi. \tag{5.40}$$

In particular, for $g(x) = 0$ this solution describes the initial wave form $f(x)$ breaking into two waves, $\frac{1}{2}f(x-ct)$ and $\frac{1}{2}f(x+ct)$, travelling in the opposite directions.

[44] To verify the 2nd term in Eq. (5.40), note that by changing the order of integration,

$$\begin{aligned}
\int_a^b g(\xi)d\xi &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\int_a^b e^{-ip\xi}d\xi G(p)dp = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\left[\frac{e^{-ip\xi}}{-ip}\right]_a^b G(p)dp \\
&= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\frac{e^{-ipa}-e^{-ipb}}{ip}G(p)dp.
\end{aligned}$$

which agrees with the last term in Eq. (5.39), for $a = x - ct$ and $b = x + ct$.